Foreign Firm Footprints: Entry, Agglomeration and Welfare in Mexico *

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Abstract

Foreign firms account for less than 1% of Mexican manufacturing establishments but 40% of employment and 60% of exports, and they agglomerate with same-origin firms into distinct regional corridors. Using data from seven waves of the Economic Census (1994–2024), this paper develops and estimates a quantitative spatial model of foreign and domestic firm location in which origin- and industry-specific agglomeration produce this geographic dispersion. I use the results to conduct counterfactual exercises evaluating how U.S. tariff impositions would affect firm location decisions, input markets, and welfare across Mexican commuting zones. Under a 25% tariff on Mexican exports, the income-weighted average commuting zone loses 0.051 log points of real income, and the losses are largest where foreign firms are most present.

Keywords: Firm Entry, Labor Markets, Offshoring, Count Data Models

JEL Codes: J42, R12, R30

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1 Introduction

Manufacturing exports are a pillar of the Mexican economy, summing to nearly a third of the country's GDP. Most of these exports are produced by foreign firms, which are few in number but large in scale. On average, a foreign plant employs orders of magnitude more workers than a Mexican-owned one, and one entrant alone can make up a large part of a local labor market. Where these firms decide to locate matters, because every plant that settles in a place makes it more attractive to later foreign entrants of its industry. Their location choices therefore shape the economic geography of the country. In this paper I document that foreign firms cluster with firms of their own origin in geographically distinct regions of Mexico, instead of concentrating in a single "foreign zone." I then ask two questions about this pattern. The first is what draws foreign entrants to a particular location. The second is how the geography that results from these choices shapes the welfare cost of a shock to exports.

Recent events have made these questions more urgent, but they have a long history. Mexico's role as an offshoring destination dates back to the 1965 Border Industrialization Program, whose *maquiladoras* imported inputs duty free to assemble goods for export along the US border. A second regime, the Temporary Import Program to Produce Export Goods (PITEX), created in 1985, extended duty-free imports to manufacturers already established in the country, Mexican and foreign-owned, that sold at home as well as abroad. The North American Free Trade Agreement (NAFTA) deepened Mexico's role as an export platform in 1994, and in 2006 the two programs were merged into the Manufacturing, Maquiladora and Export Services Industry (IMMEX) program, which grants the same benefits to any firm that exports more than US$500,000 a year or at least ten percent of its revenue, wherever its plants are located. At present, no federal policy pushes foreign investment toward any particular region. There have been place-based schemes in the past decade, but none of them shaped the geography I measure. The Special Economic Zones created in 2016 never operated and were abolished in 2019. The northern border stimulus, in force since 2019, does not give its income-tax credit to plants under the maquila tax regime. The development poles of 2023 and 2025 are too recent to matter for my data. Other than the legacy of the original border program, then, the regional footprints in this paper come from the location choices of the firms themselves, not from a policy map.

The trade war between the US and China, which started in 2018, raised the stakes of these location choices. It triggered a major reallocation of global trade, and Fajgelbaum et al. (2024) find Mexico among the countries whose exports gained. This is a notable historical reversal: Utar and Torres Ruiz (2013) show that China's accession to the World

Trade Organization in 2001 displaced substantial Mexican manufacturing activity, and for two decades China was Mexico's dominant offshoring competitor. The inflow deepened the already substantial role of foreign-owned firms in the Mexican economy: Utar (2026) reports that foreign-headquartered firms added over half a million manufacturing jobs in Mexico between 2017 and 2023, and that firms more exposed to the US tariffs on China, through their 2017 export portfolios, expanded employment faster. The US tariff policy that diverted production to Mexico has since been turned on Mexico itself. In March 2025 the United States imposed a 25% tariff on Mexican goods that did not qualify under the United States-Mexico-Canada Agreement (USMCA); after the Supreme Court struck it down in February 2026, 10% duties on the same goods took its place. These tariffs threaten exactly the export platform that nearshoring built. A large literature measures the cost of the 2018 US tariffs in the country that imposed them, down to its incidence across US counties (Amiti et al., 2019, Fajgelbaum et al., 2020), and their effect on other countries' trade (Fajgelbaum et al., 2024); Fajgelbaum and Khandelwal (2022) survey it. This paper measures the cost across the regions of the exporting country, where it depends on where the foreign exporters located.

To answer these questions I study the determinants of firm entry into Mexican labor markets, with a particular focus on the distinction between foreign and domestic firms. This distinction is important because entry does not mean the same thing for the two types of firms. A foreign entrant usually offshores production from abroad and can put its plant anywhere in the country. A domestic entrant, on the other hand, is typically a local entrepreneur who opens a plant where they already live. I develop a model in which heterogeneous firms choose locations based on profits that depend on local productivity conditions, input costs, agglomeration forces, and export access. In line with that distinction, foreign firms in the model draw a productivity in every location and choose the one where their profit is highest, while domestic firms enter their home location conditional on drawing a sufficiently high productivity. The foreign margin takes the Fréchet productivity structure of the Ricardian trade model of Eaton and Kortum (2002), which Ramondo and Rodríguez-Clare (2013), Tintelnot (2017) and Arkolakis et al. (2018) extend to multinational production across host countries and Fajgelbaum et al. (2019) to firms that draw a productivity in every US state, and applies it to foreign firms choosing among the commuting zones (henceforth CZs) of a single host country. In my model, the scale of these draws varies by origin, industry and CZ, and it depends on the local stocks of same-origin and same-industry activity. Agglomeration is therefore a primitive of the location choice. The rest of the economy follows the within-country spatial general equilibrium literature, which Caliendo et al. (2018) build on the same Fréchet structure and Allen and Arkolakis (2014) on an Armington model with spillovers.

To estimate the model I use the universe of manufacturing establishments in seven waves

of the Mexican Economic Census (1994–2024). The census identifies foreign capital participation and country of origin at the establishment level, so foreign entry can be distinguished by North American, Asian, European and Other origin groups and estimated separately from domestic entry. I estimate both margins as Poisson regressions on establishment counts. Guimarães et al. (2003) show that this is equivalent to estimating the structural location-choice probabilities, and Guimarães et al. (2004) extend the result to choices that violate the independence of irrelevant alternatives. Arauzo-Carod et al. (2010) survey the empirical literature on industrial location. The two margins correspond to the two polar cases in Schmidheiny and Brülhart (2011). A foreign entrant that is attracted to one CZ is drawn away from another, while domestic entry adds plants without taking them from anywhere else. My contribution to this literature is to estimate the two margins as a matched pair on the same census.

I find two patterns in the estimates. First, foreign entrants are strongly drawn to locations with denser same-industry clusters, and the industry-agglomeration elasticity dominates the entry estimates. A long literature finds strong agglomeration effects in the location of foreign investment (Crozet et al., 2004, Guimarães et al., 2000, Head et al., 1995); Japanese manufacturers in the US, for instance, favored states where earlier Japanese plants in their industry had settled (Head et al., 1995). In Mexico, a good part of what looks like a pull toward same-origin firms is in fact a pull toward same-industry firms. The elasticity of foreign entry to the same-origin stock drops from 0.23 to 0.06 once I include the same-industry stock. Second, the fact that each origin has its own footprint is nonetheless robust as a descriptive pattern. North American firms line the northern border, Asian firms cluster in the northwestern and central-western (Bajío) states, and European firms concentrate in the central and eastern states. The estimated origin loading is small, however. In the counterfactual the corridors therefore matter because of where firms already are, with each CZ keeping its baseline mix of origins, and not because of an estimated force that pulls in firms of the same origin.

For the counterfactual, I impose a uniform 25% tariff on all Mexican manufacturing exports. Since goods that qualified under USMCA were exempt, this is an upper bound on the tariffs actually applied. I solve for the new equilibrium with the exact-hat algebra of Dekle et al. (2008), which Caliendo and Parro (2015) also use in their quantification of NAFTA. Welfare changes are decomposed with standard CES price-index accounting into an income channel and a price channel, and the price channel is in turn split into a variety term, as in Krugman (1980), and a productivity and input-price term, as in Eaton and Kortum (2002). I find two main results. *First, the loss is large, and every CZ loses under the all-exporter tariff.* Weighted by income, the average CZ loses 0.042 log points of income, and 0.051 log points of welfare once I add the increase in the tradable price index. If only foreign-owned exporters

pay the tariff, the income loss is 0.018, which is still over two-fifths of the loss under the blanket tariff. Most of the income loss comes through wages (-0.038, compared to -0.003 for domestic profits). The local wage clears against every local employer, so it moves by the tradable share of local labor payments, which is about a third on average. Domestic profits fall little because the entry of Mexican-owned plants cushions them. The price channel adds another 0.010 to the loss. Fewer plants raise the tradable price index, cheaper inputs offset two-fifths of this increase, and productive capacity hardly changes.

Second, the loss is concentrated where foreign exporters operate. Across CZs, the welfare change has a standard deviation of 0.033 under the all-exporter tariff and 0.017 under the foreign-only tariff, and 63 of the 738 CZs in the counterfactual panel lose more than 0.10 log points. This variation comes from the income channel and follows the geography of foreign firms that I document in Section 2. CZs with any foreign presence at baseline hold 96% of national income. Where foreign plants earn more than a fifth of tradable revenue, welfare falls by 0.079, compared with 0.021 in CZs with no foreign firms, and these CZs bear three-quarters of the aggregate loss. The largest losses are along the northern border, where foreign plants hold almost half of tradable revenue and tradables pay more than half of the wage bill, and in Bajío centers such as Aguascalientes and Querétaro. The central-eastern states, where the European corridor is, lose less than the national average. Since foreign exporters of every origin are spread over several distinct corridors, one tariff hits all of them at the same time. Within each CZ, it is the local labor market that carries the shock from the exporters to the rest of the CZ. Entry links a plant to its neighbors: large plant openings raise the productivity of incumbent plants (Greenstone et al., 2010), and jobs created by foreign multinationals raise employment and value added at domestic firms in the same commuting zone (Setzler and Tintelnot, 2021). My model has both kinds of links, the shared local wage and the agglomeration stocks. Under the tariff, however, it is the wage that matters. Through it, the export loss of foreign plants reaches every worker in the CZ, while the estimated agglomeration feedback adds almost nothing to the loss (Section 5.4).

The remainder of this paper is organized as follows. Section 2 documents three stylized facts about foreign-owned manufacturing in Mexico that motivate the analysis. Section 3 develops the theoretical framework. Section 4 describes the data, the empirical strategy and the estimates. Section 5 presents the counterfactual analysis, and Section 6 concludes.

2 Stylized Facts

This section documents three stylized facts about foreign-owned manufacturing in Mexico that motivate the model and empirical strategy developed in the rest of the paper: (i) foreign

firms are few but large and export-oriented, (ii) their geographic footprint varies sharply by region of origin, and (iii) foreign firms of every origin concentrate in the same few export industries, led by transportation equipment, which hold only a quarter of the Mexican-owned wage bill.

Throughout the paper I focus on the tradable sectors, that is, manufacturing (NAICS 31–33) and non-petroleum mining (NAICS 212–213). Together they account for most Mexican exporters, and they are the natural setting for the trade-cost mechanism of the model. I leave out oil and gas extraction (NAICS 211), since in Mexico it is dominated by state-owned firms and works under a regulatory regime that cannot be compared with the rest of the tradable sector. For brevity, I refer to this tradable sample as "manufacturing" throughout the paper, including in Table 1.

2.1 The Scale of Foreign Presence

Foreign-owned establishments make up less than 1% of Mexican manufacturing plants in 2024 (5,066 foreign against 619,674 Mexican-owned, Table 1), yet they account for a disproportionate share of overall activity. Table 1 compares aggregate totals for Mexican-owned and foreign-owned establishments across three labor measures (employment, total occupied labor and the wage bill) and two revenue measures (total and export), and the change in the foreign share of each since 1994.

Table 1: Foreign vs. Mexican-Owned Footprint in Mexican Tradable Industries, 2024

	Mexico	North America	Asia	Europe	Other	Foreign Share	Change since 1994
Employment (thousands)	3,556	1,734	312	650	123	0.442	+0.182
Total occupied labor (thousands)	4,348	1,734	312	650	123	0.393	+0.163
Wage bill (billion 2023 pesos)	375	160	27	61	11	0.409	+0.166
Total revenue (billion 2023 pesos)	2,159	568	103	248	47	0.309	+0.141
Export revenue (billion 2023 pesos)	325	303	53	136	21	0.613	+0.333[†]
Total establishments	619,674	3,183	476	1,175	232	0.008	−0.001

Source: Author's calculation from the 1994–2024 Economic Censuses, INEGI. Change: foreign share in 2024 minus foreign share in 1994; [†]in 1999. Revenue and wage bill winsorized at 1% level.

Foreign-owned establishments employ 44% of manufacturing workers on the payroll, pay out 41% of the manufacturing wage bill, and generate 31% of total manufacturing revenue. Their share of total occupied labor (which also counts unpaid workers) is lower at 39%, because Mexican firms include a meaningful mass of unpaid workers in their labor rolls while foreign firms rely almost entirely on paid employees, so their total occupied labor is close to their employment. The wage-bill share (41%) is well above the revenue share (31%) because foreign plants are much larger and spend more of their revenue on labor. On average, a foreign plant has 557 paid workers, against 5.7 for a Mexican-owned plant. The foreign footprint is

largest on the export side, where foreign firms account for 61% of Mexican manufacturing export revenue.

Most of this gap comes from North American firms. On their own they generate 303 billion 2023 pesos in export revenue, which is roughly as much as the whole Mexican-owned sector (325 billion pesos) and 36% of all Mexican manufacturing exports. European firms account for another 16%, Asian firms for 6% and firms of Other origins for 2%. The disproportionate scale of foreign presence in Mexican manufacturing is therefore concentrated on the export side and, within it, on North American firms.

2.2 Geographic Heterogeneity by Origin

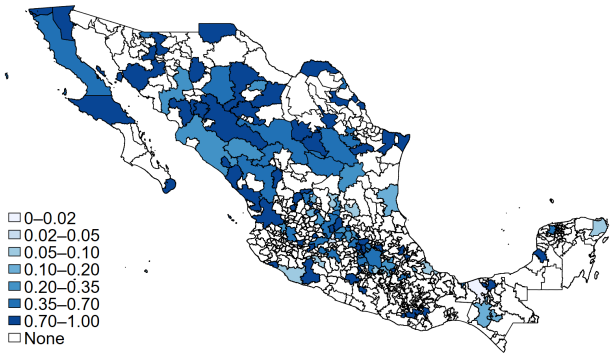
Figure 1 maps the origin composition of the foreign wage bill across Mexican commuting zones in 2024. The color of a CZ in each panel is the share of its *foreign* wage bill paid by plants of the given origin. The maps therefore show how foreign activity is divided among origins, but they say nothing about how much foreign activity a CZ has. The patterns are quite different across the three origins. North American firms dominate in the northern border states. This goes back to the deep integration with US supply chains that has defined the *maquiladora* industry since the 1960s. Asian firms cluster predominantly in the northwestern and central-western (Bajío) states, where Japanese, Korean and Chinese auto and electronics manufacturers arrived in more recent waves of investment. European firms follow a third pattern and are more present in the central and eastern regions of the country.

This dependence of location on the region of origin, beyond the usual cost and market-access fundamentals, is the main reason I include origin-specific agglomeration as a primitive of the model. Take North American firms near the US border. Proximity to the US is one reason they are there, but decades of supply-chain integration have also left behind networks of suppliers, workers and infrastructure that make the border cheaper to operate in for firms from the same origin. In the model I capture this by making a firm's productivity in a location depend on the presence of firms of its own origin in that commuting zone and, through a separate channel, on the presence of firms of its own industry.

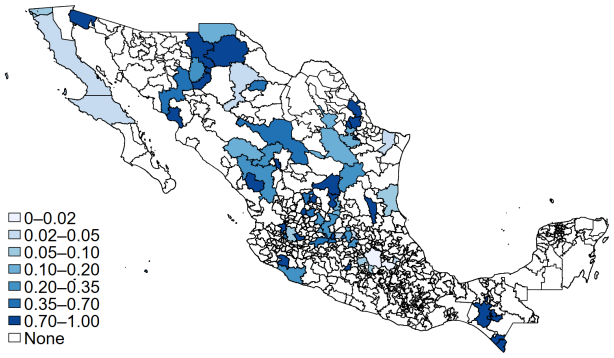
2.3 A Shared Export Core

The origin-differentiated corridors of Figure 1 do not come with origin-differentiated industries. Foreign firms of every origin concentrate in the same export core: the five industries that lead the foreign wage bill over 1994–2024, transportation equipment (NAICS 336), computer and electronic products (334), plastics and rubber (326), electrical equipment (335) and chemicals (325). Figure 2 splits each group's 2024 wage bill between these five industries

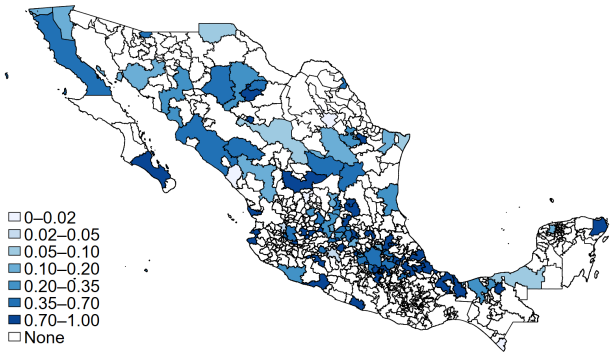
Figure 1: Origin Composition of the Foreign Wage Bill by CZ, 2024



(a) North America



(b) Asia

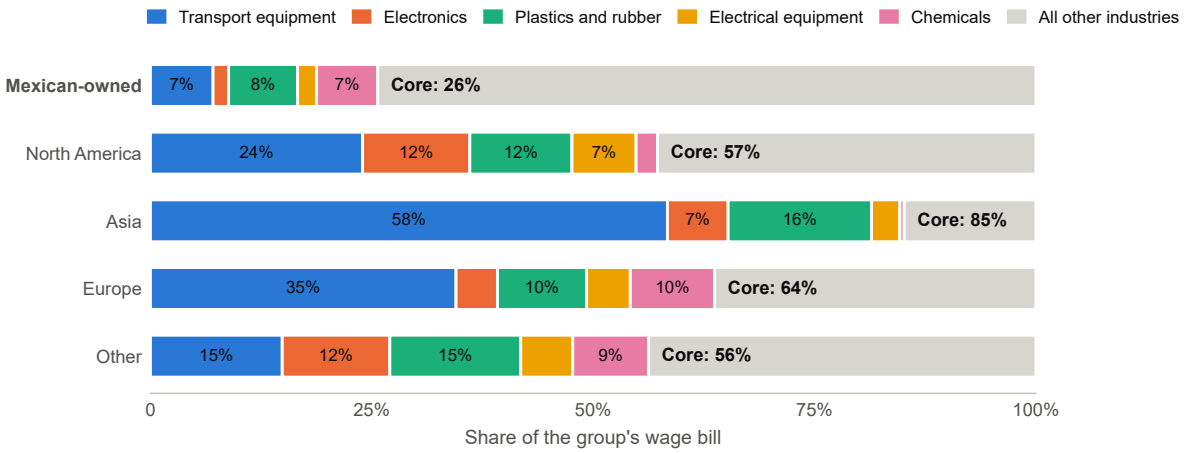


(c) Europe

Note: Share of the foreign wage bill paid by plants of the indicated origin (Mexican-owned plants excluded). White: no plants of that origin in the CZ. Source: 2024 Economic Census, INEGI.

and the rest. The core holds between 56% (Other) and 85% (Asia) of each foreign group's wage bill, against 26% of the Mexican-owned wage bill, whose largest industry, food (22%), holds between 1% (Asia) and 9% (Other) of the foreign ones. Where the groups differ is in how their wage bill is split within the core. Asian firms have 58% of their wage bill in transportation equipment alone, European firms 10% in chemicals and North American firms 12% in electronics.

Figure 2: A Shared Export Core, 2024



Note: Share of each group's 2024 wage bill in the five industries that lead the foreign wage bill over 1994–2024. Source: Economic Censuses, INEGI.

This shared core is important for reading the entry estimates. Foreign firms of every origin are in the same few industries, and when a foreign group is present in a given industry and CZ, it only has a handful of plants there, between 1.5 (Other) and 3.9 (North America) on average, against 53 for Mexican-owned plants. In other words, the plants of a foreign group in a CZ are mostly in a few core industries. As a consequence, within a CZ the same-origin and same-industry agglomeration measures are nearly collinear, which turns out to matter for the origin loading in Section 4.3. At the same time, the groups that settle in different regions mostly work in the same industries, so industry mix by itself cannot produce the corridors of Figure 1.

Each of these facts shapes the model. Because of fact (i), I model foreign and domestic firms differently. A foreign firm runs a large, export-oriented plant, picks among commuting zones, and weighs the option to export when it does so. The typical Mexican-owned plant is small, opens where its owner already is, and enters only if its productivity is above a cutoff. Fact (ii) means the model needs an agglomeration force tied to the origin of the firm, and fact (iii) means it also needs a separate one tied to its industry. Tradable plants also hire from the same local pool as every other employer, so the wage has to clear the local labor market as a whole, non-tradables included. A tariff on exports then hits foreign-firm cells

disproportionately, but it also reaches domestic-firm cells through the local wage and the input bundle, and changes domestic entry that way (Section 5). In Section 3 I build a model with these properties.

3 Model

The facts of Section 2 call for five ingredients. First, I need heterogeneous productivity, since only some firms enter and only some export. Second, foreign firms have to choose a location, while domestic firms enter where they are. Third, how productive a location is should depend on who is already there, both by origin and by industry. Fourth, input prices are set locally, so entry affects the wage and the wage affects entry. And fifth, there is an export market that firms reach at a trade cost, which is how the tariff comes in. I combine these ingredients in a location model with Fréchet productivity and monopolistic competition. Following the discrete-choice structure of Eaton and Kortum (2002), foreign firms draw a productivity in every CZ and locate where their profit is the highest. I start with the consumer's problem, which gives demand and welfare. I then turn to the two firm problems and close the section with the equilibrium.

3.1 Consumer's Problem

Each commuting zone c hosts a representative household, with preferences identical across CZs, that allocates expenditure between a tradable bundle C_M and a non-tradable bundle C_S according to Cobb-Douglas preferences with tradable share $\chi \in (0,1)$. Within tradables, the household aggregates industry sub-bundles C_j via Cobb-Douglas expenditure shares α_j (summing to one), and within each industry j has CES preferences over a continuum of differentiated varieties produced by monopolistically competitive firms. The household supplies its labor inelastically in its own CZ. Each location thus has a fixed labor supply L_c, which is divided between tradable and non-tradable employment at a common local wage w_c. Utility is

$$U = \max_{\{q(\omega)\}} C_M^{\chi} C_S^{1-\chi}, \qquad C_M = \prod_{j=1}^{J} C_j^{\alpha_j} \tag{1}$$

subject to

$$C_j = \left(\int_{\omega \in \Omega_j} q(\omega)^{\frac{\sigma-1}{\sigma}} \, d\omega \right)^{\frac{\sigma}{\sigma-1}}, \qquad \sum_{j=1}^{J} \alpha_j = 1,$$

$$\sum_{j=1}^{J} \int_{\omega \in \Omega_j} p(\omega)\, q(\omega)\, d\omega + P_S C_S \;\leq\; w_c L_c + \Pi_c^d,$$

where quantities are those of the household in c, whose index I suppress.

Here, χ is the share of household expenditure on tradables, α_j are the within-tradable expenditure shares, $\sigma > 1$ is the elasticity of substitution between varieties, and w_c denotes the local wage. The household of CZ c owns the domestic tradable firms located in c, so its income includes dividends Π_c^d in addition to labor earnings $w_c L_c$ (aggregating tradable and non-tradable wages under the common local wage).

The non-tradable sector is competitive and produces under constant returns to scale, so its producers price at cost and earn zero profits; it is not modeled in further detail. Its good serves as the numeraire, $P_S \equiv 1$, so wages, tradable prices and income are all measured in units of the non-tradable good.[1] The two-tier structure leaves the firm side, entry, and tradable prices unchanged relative to a one-sector formulation. The tradable sector is broader than manufacturing: it also includes mining. Its industries form the set $\mathcal{J}$, which the equilibrium block (Section 3.3) and the counterfactual also write as $\mathcal{T}$, with N for the non-tradable sector.

3.2 Firm's Problem

Firms are heterogeneous across four dimensions: their industry $j \in \mathcal{J}$, location $c \in \mathcal{C}$ (the set of commuting zones), origin-specific core productivity $\phi_o \sim G_o(\phi)$, and location-specific productivity shifter $z_{ojc} \sim F_{ojc}(z)$, drawn independently of ϕ_o. For simplicity, I drop the time subscript in this section.

Foreign firms ($o \neq d$) and domestic firms ($o = d$) share this heterogeneity structure but differ in one central respect: foreign firms choose, among Mexican commuting zones, the one where their value is highest, whereas domestic firms operate at a fixed location and enter conditional on drawing productivity above an entry cutoff.

Before making its choice, a firm knows its core productivity ϕ_o and draws its location-specific shifter z_{ojc}. A foreign firm gets one draw for every CZ and compares them, whereas a domestic firm only draws once, at home. Realized productivity is then given by $\phi_o z_{ojc}$.

[1] In the wage-pinning closure of Antràs et al. (2017), a constant-returns, labor-only numeraire sector fixes the wage. Here w_c is determined by local labor-market clearing in (11), and $P_S \equiv 1$ is a choice of units in each CZ.

Following the literature, I assume that z_{ojc} is drawn from a Fréchet distribution with scale parameter $T_{ojc} > 0$ and shape parameter $\theta > \sigma - 1$:

$$F_{ojc}(z) = \exp\big(- T_{ojc}\, z^{-\theta}\big).$$

The scale parameter captures the absolute advantage of a location c for firms of origin o operating in industry j: a higher T_{ojc} shifts the distribution of z_{ojc} upward for all firms in that subgroup, raising realized productivity $\phi_o z_{ojc}$ accordingly. Agglomeration comes in through this scale, $T_{ojc} = \exp(\beta' X_{ojc})$, where the vector X_{ojc} contains the same-origin and same-industry wage-bill stocks in c. The shape parameter, in turn, governs how dispersed the productivity draws are across the plants of a cell.

A firm sells domestically and has the option to export. In order to sell to a market $m \in \{d, x\}$, the firm faces an iceberg trade cost $\tau^m_{jc} \geq 1$: the firm must ship τ^m_{jc} physical units to deliver one unit at market m, so the wedge $\tau^m_{jc} - 1$ captures all per-unit costs of moving goods from location c to the destination: tariffs, freight, customs frictions, and other border barriers. Domestic trade is assumed costless, $\tau^d_{jc} = 1$, so the wedge applies only on the export side as τ^x_{jc}. The firm also faces an input-price bundle $\tilde{P}_{jc}$, a Cobb-Douglas aggregate over inputs indexed by k:

$$\tilde{P}_{jc} = \prod_k p_{kc}^{\beta_{kj}}; \quad \sum_k \beta_{kj} = 1,$$

where β_{kj} is the cost share of input k. Empirically the bundle is labor, materials and energy, and only the wage is set locally: materials are priced in national and international markets (Mexico is a small open economy) and energy prices are regulated, so $\tilde{P}_{jc}$ varies across CZs through w_c alone (Appendix A). Capital enters through the fixed cost, not the variable bundle. Under CES demand and monopolistic competition, standard pricing yields

$$p^m(\phi_o z_{ojc}) = \frac{\sigma}{\sigma - 1} \frac{\tau^m_{jc} \tilde{P}_{jc}}{\phi_o z_{ojc}}. \tag{2}$$

Aggregating (2) over all varieties sold in the domestic market, produced in every location c, gives the industry price index. This index carries the whole price side of the welfare analysis: entry, productivity and input prices reach consumers through it, and, because domestic trade is costless, it is common to every commuting zone:

$$P^d_j = \left[\int_{\omega \in \Omega_j} p(\omega)^{1-\sigma}\, d\omega \right]^{\frac{1}{1-\sigma}}. \tag{3}$$

Given this pricing rule, the firm's revenue in market m follows directly from CES demand.

Substituting (2) into the demand system as in (1) yields revenue for each market m,

$$r^m(\phi_o z_{ojc}) = \left(\frac{\sigma}{\sigma - 1} \frac{\tau_{jc}^m \tilde{P}_{jc}}{\phi_o z_{ojc}} \right)^{1-\sigma} Y_j^m \left(P_j^m \right)^{\sigma-1},$$

where Y_j^m is expenditure on industry j in market m and P_j^m the corresponding industry price index; in the domestic market $Y_j^d \equiv \chi \alpha_j Y_d$, with Y_d national income from (15), and P_j^d is (3). Piecing everything together, the firm's profits from each market m are given by

$$\pi^m(\phi_o z_{ojc}) = \frac{1}{\sigma} r^m(\phi_o z_{ojc}) - f_c^m, \tag{4}$$

where f_c^m is the fixed cost of serving market m from location c: a requirement common to all industries in c, paid in nationally priced inputs rather than local labor, so it is a constant in units of the numeraire. It is paid every period in which the firm serves market m; there is no sunk entry cost and no free-entry condition. The firm selects into exporting only if z_{ojc} is sufficiently high to cover the additional fixed costs of serving the export market in addition to already serving the domestic market. Thus total profits are given by

$$\Pi(\phi_o z_{ojc}) = \pi^d(\phi_o z_{ojc}) + \mathbb{1}_{ojc}^x \pi^x(\phi_o z_{ojc}). \tag{5}$$

Here $\mathbb{1}_{ojc}^x$ is an indicator equal to one if the firm is productive enough to serve the export market. I replace it with the cell's export propensity $p_{ojc}^x \equiv \mathbb{P}(z_{ojc} \geq z_{ojc}^{x,*})$, the share of the cell's plants that clear the threshold at which export profits cover the export fixed cost, and take it as given, without solving for the threshold. Since each market's revenue scales with $(\tau_{jc}^m \tilde{P}_{jc})^{1-\sigma} (\phi_o z_{ojc})^{\sigma-1}$, total revenue is then domestic revenue times $1 + p_{ojc}^x \tau_{jc}^{x,1-\sigma}$, up to export-market demand terms common to all CZs: the unity captures the always-available domestic market and the $p_{ojc}^x \tau_{jc}^{x,1-\sigma}$ term the additional gain from the export option scaled by the iceberg trade cost.

Foreign firms. I embed the Eaton and Kortum (2002) structure into the problem of a firm deciding where to establish a single plant. The problem is close to that of Tintelnot (2017), with a different timing: his firms choose a set of plants before they observe their plant-level efficiencies, while here a firm sees its draw in every CZ and then chooses. What matters for the firm is that its profits in a location depend on its own productivity and also on the costs and demand conditions of that location. There is a potential mass of entrants M_{oj}^E of origin o in industry j. Each potential entrant draws its location-specific productivity z_{ojc} for every CZ, independently across CZs and firms, and knows its core productivity ϕ_o, which scales every location's value equally and so does not affect where it goes; what sends firms of the

same origin and industry to different CZs are their draws. The entrant chooses the location with the highest value:

$$v(\phi_o) = \max_{c \in \mathcal{C}} \left\{ 0,\ \sum_{t=0}^{\infty} (1 - \delta_{jc})^t \, \Pi(\phi_o z_{ojc}) \right\}. \tag{6}$$

After entry, the firm pays the per-period fixed cost f_c^d of serving the domestic market and stays in c indefinitely, but exits each period with probability δ_{jc}, an exogenous exit hazard that varies across industries and CZs with conditions the model does not track, such as local security, institutions and industry churn. Since there is no discounting other than exit, the value of a location equals its per-period profit divided by the exit rate. A foreign plant only operates if its per-period profit in the chosen CZ, from both domestic and export sales, covers its fixed costs. In other words, its draw in that CZ has to clear the cutoff $z_{ojc}^{f,*}$ at which this profit is zero, where the export term is summarized by p_{ojc}^x as before. Firms with draws below the cutoff stay out, and plants whose profit drops below their fixed costs exit. This is why a lower return on exports raises the cutoff.

The foreign firm sorts into the location whose value is highest. The probability that CZ c is chosen is increasing in the location's productivity scale T_{ojc}, decreasing in its input costs $\tilde{P}_{jc}$ and its exit rate δ_{jc}, and increasing in its export profitability, captured through p_{ojc}^x and τ_{jc}^x:

$$\mathbb{P}(c \mid oj) = \mathbb{P}\left(\frac{\Pi(\phi_o z_{ojc})}{\delta_{jc}} \ \geq \ \frac{\Pi(\phi_o z_{ojc'})}{\delta_{jc'}} \quad \forall\, c' \in \mathcal{C} \mid oj \right) \tag{7}$$

The optimal location choice for a foreign firm is given by:

$$c^* = \underset{c \in \mathcal{C}}{\operatorname{argmax}} \ \frac{\Pi(\phi_o z_{ojc})}{\delta_{jc}} \tag{8}$$

Given that z_{ojc} is continuous and independent across CZs, the probability that two locations yield the same value is zero. Therefore, the firm will always have a unique optimal location choice.

At the location stage a CZ's value is proportional to $z_{ojc}^{\sigma-1} \tilde{P}_{jc}^{1-\sigma} (1 + p_{ojc}^x \tau_{jc}^{x,1-\sigma}) / \delta_{jc}$, and $z_{ojc}^{\sigma-1}$ is Fréchet with shape $\theta/(\sigma-1)$ and scale T_{ojc}, so the profits from locating in c are proportional to

$$\frac{\Pi(\phi_o z_{ojc})}{\delta_{jc}} \ \propto \ \phi_o^{\theta} \, \delta_{jc}^{-\frac{\theta}{\sigma-1}} \, T_{ojc} \, \tilde{P}_{jc}^{-\theta} \left(1 + p_{ojc}^x \, \tau_{jc}^{x,1-\sigma}\right)^{\frac{\theta}{\sigma-1}}. \tag{9}$$

Expression (9) is what the firm's location choice in (8) ranks across commuting zones, and it is the object that enters the Poisson regression in the empirical section: (7) has the closed form in which the probability that c is chosen is the right-hand side of (9) for c over its sum

across CZs, from which ϕ_o cancels. Note that T_{ojc} enters the choice probability one for one, since it is the scale of the draws, while every other factor of the value, including the exit rate, is raised to the power $\theta/(\sigma-1)$. The closed form also leaves out fixed costs at the location stage. It is exact if fixed costs are zero or proportional to variable profit, and an approximation otherwise. Fixed costs matter once the location is chosen, where they set the cutoff for operating.

Domestic firms. Domestic firms are assumed to have a fixed location c and maximize the value of operating in that location. The timing of their decision follows the traditional entry timing of Hopenhayn (1992). Each jc pair has a mass of potential entrants M^E_{djc}. An entrant learns its location-specific productivity draw z_{djc} and operates only if its per-period profit, domestic and export, covers its fixed costs, that is, if the draw clears the cutoff $z^{d,*}_{jc}$ at which that profit is zero, the same rule as for foreign plants. Operating firms exit at the same rate δ_{jc} as foreign firms. Their value function is

$$v(\phi_d) = \max\left\{0,\ \sum_{t=0}^{\infty}(1-\delta_{jc})^t\,\Pi(\phi_d z_{djc})\right\}. \tag{10}$$

3.3 Equilibrium

The labor market clears in each location c; the nationally set prices of energy and materials enter $\tilde{P}_{jc}$ as constants common to all CZs. Factor markets are competitive, so every input is paid its share of cost: a share of revenue net of the markup $\sigma/(\sigma-1)$ in tradables, and of revenue in non-tradables, which price at cost (Appendix A). Workers are perfectly mobile within each commuting zone, so the wage w_c is the same for *all* local employers, tradable or not. Manufacturing, mining, wholesale and retail trade, transport, construction and services all hire from one local pool and pay the same wage. Writing $\mathcal{T}$ for the tradable industries the model follows individually and N for the non-tradable remainder, and imposing $\sum_{j\in\mathcal{T}} L_{jc} + L_{Nc} = L_c$,

$$w_c \;=\; \frac{\frac{\sigma-1}{\sigma}\sum_{j\in\mathcal{T}}\beta_{Lj}\,R_{jc} \;+\; \beta_N\,R_{Nc}}{L_c}, \tag{11}$$

where R_{jc} is the revenue of tradable industry j in c and β_{Lj} its labor cost share, and R_{Nc} is non-tradable revenue in the commuting zone and β_N its labor cost share. Total employment L_c in the CZ does not change, but workers move between the two sectors as relative revenues change. Note that the non-tradable term is what keeps the pass-through below one. Without it, a tradable shock would move the local wage one for one, and with it the pass-through is the tradable share of local labor payments.

As for output prices, using the properties of the Fréchet distribution and the independence between ϕ_o and z_{ojc}, one can express the aggregate industry-level price index (3) as:

$$P_j^d = \left(\frac{\sigma}{\sigma-1}\right)\left(\sum_o M_{jo}\,\mathbb{E}[\phi_o^{\sigma-1}]\,\Gamma\left(1-\frac{\sigma-1}{\theta}\right)\right)^{\frac{1}{1-\sigma}}\left(\sum_c T_{jc}\tilde{P}_{jc}^{-\theta}\right)^{-\frac{1}{\theta}} \tag{12}$$

where M_{jo} is the mass of firms of origin o in industry j, so that $M_j \equiv \sum_o M_{jo} = \sum_c M_{jc}$ is the industry's total mass, and $T_{jc} \equiv \sum_o T_{ojc}$ is the origin-aggregated scale parameter for industry j in location c. Γ denotes the gamma function, which arises from the Fréchet expectation. Equation (12) shows that more productive firms, a larger mass of firms and lower input prices all lower the price index.

Given the households' two-tier utility in (1) and prices P_j^d, the tradable price index and the aggregate consumer price index are

$$P^M = \prod_{j=1}^{J}\left(P_j^d\right)^{\alpha_j}, \qquad P = \left(P^M\right)^{\chi} P_S^{1-\chi}. \tag{13}$$

With the non-tradable good as numeraire, $P_S \equiv 1$, the aggregate index collapses to $P = (P^M)^{\chi}$: only P^M responds to tradable entry and to the tariff shocks studied below, and it enters at weight χ through the outer Cobb-Douglas aggregator (both indices omit the Cobb-Douglas constants, which drop out of every log change below).

Having characterized the prices in the economy, I can now define welfare. Households supply their labor inelastically to the commuting zone as a whole, and they earn the common local wage w_c whether they work in a tradable or in a non-tradable industry. They also receive the dividends of the domestic tradable firms located in their CZ, while non-tradable producers make no profits. Household income thus includes the *total* local wage bill $w_c L_c$, and not just the part paid by tradables. Welfare in commuting zone c is its total real income:

$$W_c = \frac{w_c L_c + \Pi_c^d}{P}. \tag{14}$$

Since L_c is fixed, real income per worker W_c/L_c moves one for one with W_c, so its log changes are also per-capita changes. Domestic income Y_d closes the model. Summing over CZs, national income is the sum of CZ-level labor earnings and domestic dividends:

$$Y_d = \sum_{c\in\mathcal{C}}(w_c L_c + \Pi_c^d). \tag{15}$$

I treat the export-market price index P_j^x and expenditure Y_j^x as exogenous and hold them fixed throughout; the export tariff of Section 5 acts through the export trade cost τ_{jc}^x, not

through P_j^x or Y_j^x. Now that I have defined the different components of the economy, I can formally define the equilibrium of the model:

Definition 1 (Equilibrium)**.** Given τ_{jc}^m, f_c^m, p_{ojc}^x, δ_{jc}, the scales T_{ojc}, the shape θ and loadings β, the potential-entrant masses M_{oj}^E and M_{djc}^E, $\forall\, c \in \mathcal{C}$, $G_o(\phi)$, and the export-market aggregates P_j^x and Y_j^x, an equilibrium is a set of wages w_c, input prices $\tilde{P}_{jc}$, domestic price indices P_j^d, national income Y_d, firm masses M_{ojc}^f and M_{jc}^d, allocations $q(\omega)$ to the representative household of each CZ, prices $p^m(\phi_o z_{ojc})$ and location choices for foreign firms such that:

- Each CZ's household maximizes utility according to (1) taking prices and wages as given.
- (8) solves the foreign firm's location choice problem in (6), whereas domestic firms enter according to (10).
- The labor market clears in each location c according to (11).
- Price indices for each industry $j \in \mathcal{J}$ satisfy (12), and aggregate price indices satisfy (13).
- Domestic income Y_d satisfies (15).

4 Empirics

This section describes the data, the empirical strategy for each type of firm, the recovery of the structural parameters, and the estimates.

4.1 Data

The primary data source is the Mexican Economic Census of 1994 to 2024, collected every five years by the National Institute of Statistics and Geography (INEGI). It covers every establishment in the Mexican economy with a fixed location, so it excludes ambulant vendors, and each establishment reports on its operations in the calendar year before the census. Restricting the sample to tradable industries, I observe each establishment's revenue, output and the inputs used in production: the number of workers, the annual payroll, and the annual expenditure on materials and energy (electricity, gas and fuel).

The census also records each establishment's industry, the year it started operating and its foreign capital participation. Establishments report whether foreign capital participates in their share capital and, if so, the country it primarily comes from and the percentage of

total capital that comes from abroad. Foreign capital here is the contribution of resources in money or kind received from abroad, not the value of assets. I define a foreign-owned establishment as one reporting at least 10% of its capital from abroad, the direct-investment threshold of the OECD and IMF benchmark definitions; the overwhelming majority of these establishments report a far larger share (Appendix C). Origin groups are Mexico, North America (US and Canada), Asia, Europe, and Other.

Local labor markets are the commuting zones of Blyde et al. (2023), which group municipalities by commuting flows. Within a commuting zone, workers can change employers without having to move, and this is why I treat it as the unit at which there is a single local wage, the wage that clears the labor market in (11). Industries are the 3-digit NAICS sectors of the tradable sample. Distances to the nearest US border crossing, seaport and airport handling exports of the same industry are built from SAT customs declarations (*pedimentos*), available from 2005. Since 2009 INEGI has provided a unique establishment identifier that links census waves; before 2009 I rely on the concordance tables of Busso et al. (2018).

Table 2 reports pooled summary statistics for the variables entering the two entry equations, at the (origin group, industry, CZ) cell level for the foreign side and the (industry, CZ) or industry level for the rest.

Table 2: Summary Statistics of the Estimation Variables, Pooled 1999–2024

	Mean	SD	p25	Median	p75	N
Foreign entrants, new plants	0.50	1.08	0.00	0.07	0.55	6,759
Foreign entrants, acquisitions	0.45	0.96	0.00	0.00	0.59	6,759
Domestic entrants, new plants	20.78	157.24	0.63	2.15	8.78	50,221
Domestic potential entrants M^E_{djct}	287.9	1,881.7	1.61	10.42	64.83	50,744
Same-origin share of CZ wage bill, $t-5$	0.21	0.19	0.07	0.15	0.33	6,759
Same-industry share of CZ wage bill, $t-5$	0.08	0.18	0.00	0.01	0.05	50,744
Export orientation $\omega^x_{ojc,t-5}$	0.05	0.13	0.00	0.02	0.07	56,980
Exit rate δ_{jc}	0.34	0.17	0.24	0.36	0.46	50,744
Labor cost share $\beta_{Lj,t}$	0.31	0.07	0.27	0.29	0.33	132
Birth rate b_{jt}	0.57	0.13	0.48	0.56	0.66	137

Note: Pooled 1999–2024, count-weighted; quantiles average the wave quantiles. Entrants have no prior plant in the CZ; foreign rows pool the four foreign origins. ω^x pools foreign and Mexican-owned cells. Agglomeration rows are shares of the lagged CZ wage bill; the regressors are $\ln(1+wL)$ of the lagged level stocks (thousands of 2023 pesos).

The non-tradable block. The wage equation (11) needs non-tradable labor payments $\beta_N R_{Nc}$. In the model labor payments are the wage bill, so both objects the paper uses are measured directly from census wage bills over every fixed establishment in the country, not

only the industries the model follows: the CZ wage w_c is the wage bill over paid employment, and the tradable share of local labor payments λ_c is the tradable wage bill over the total (Section 5). Establishments with no paid workers, mostly micro-establishments run by proprietors and family members, report no wage bill, so they add nothing to w_c or λ_c. In the tradable sector their revenue still enters the dividends Π_c^d, but the labor income of unpaid workers is counted neither there nor in the wage bill, so household income $w_c L_c + \Pi_c^d$ excludes it.

4.2 Empirical Approach

Because foreign and domestic firms solve different problems in the model, I follow a different empirical strategy for each.

Foreign firms. Under the Fréchet draws of Section 3, the choice probability (7) is expression (9) for c over its sum across CZs, from which ϕ_o cancels:

$$\mathbb{P}(c \mid ojt) = \frac{\delta_{jc}^{-\frac{\theta}{\sigma-1}} T_{ojc} \tilde{P}_{jc}^{-\theta} (1 + p_{ojc}^{x} \tau_{jc}^{x,1-\sigma})^{\frac{\theta}{\sigma-1}}}{\sum_{c'} \delta_{jc'}^{-\frac{\theta}{\sigma-1}} T_{ojc'} \tilde{P}_{jc'}^{-\theta} (1 + p_{ojc'}^{x} \tau_{jc'}^{x,1-\sigma})^{\frac{\theta}{\sigma-1}}}. \tag{16}$$

To parameterize the observables in this estimation I follow the approach of Head and Mayer (2019) and adopt the following functional forms:

$$T_{ojct} = \exp(\beta' X_{ojct}), \qquad (1 + p_{ojct}^{x}\, \tau_{jc}^{x,1-\sigma}) = \exp(\gamma' Z_{ojct}).^{2}$$

Duranton and Puga (2004) review the channels through which agglomeration enhances productivity: input sharing, labor pooling, learning, and matching between firms and workers. To capture these effects through the scale parameter T_{ojc}, the vector X_{ojct} includes two agglomeration regressors defined at the commuting-zone level: an *origin* measure summing the wage bill of same-origin firms in CZ c across all tradable industries, and an *industry* measure summing the wage bill of same-industry firms in CZ c across all origins, both rolling historical averages measured in thousands of 2023 pesos. Both are entered in $\ln(1+\cdot)$ form: writing $wL_{t-5}^{o,c}$ and $wL_{t-5}^{j,c}$ for the two sums,

$$\text{origin-agglom}_{ojct} \;=\; \ln\!\big(1 + wL_{t-5}^{o,c}\big), \qquad \text{industry-agglom}_{ojct} \;=\; \ln\!\big(1 + wL_{t-5}^{j,c}\big).$$

The $\ln(1+\cdot)$ accommodates the many cells with zero same-origin (or same-industry) labor. I

[2] First-order Taylor approximation in p_{ojct}^{x} and $\ln \tau_{jc}^{x}$.

measure the stocks with the wage bill so that each plant counts according to its size and pay, which are the margins on which labor pooling and matching operate. With this measure, a large cluster that pays high wages counts for more than many small plants.

The agglomeration measures and the export-orientation regressor ω^x_{ojct} are not entered as single one-wave lags: each is the expanding mean of the lagged stock over the census waves up to t within its cell, entered inside the $\ln(1+\cdot)$, so the regressor summarizes the cell's accumulated presence rather than a one-period snapshot.[3] Export orientation is measured by the export share of revenue ω^x_{ojct}, the historical average of lagged export-revenue shares in the ojc subgroup, which in the model rises with the export propensity p^x_{ojct}; it enters both as a standalone regressor (capturing the persistent component of an ojc cell's export orientation) and as an interaction with a vector of log distances between the commuting zone and the nearest US border crossing, maritime port, and airport reporting exports of the same industry.[4]

The number of observed entrants from origin o in industry j and location c at time t is m_{ojct}: plants whose reported start-of-operations year falls between census waves, plus acquisitions (plants linked across waves $t-5$ and t by the establishment identifier whose foreign participation rises from below to at least 10% with owners of origin o); both are counted only for firms with no prior establishment in the CZ, so it is entry, not expansion. Given the mass of potential entrants M^E_{ojt}, these sort into different locations according to:

$$m_{ojct} = M^E_{ojt} \times \mathbb{P}(c \mid ojt).$$

The pool M^E_{ojt} of potential entrants of origin o in industry j is proxied by the pair's lagged foreign labor stock; it varies only at the origin-industry-year level. In order to estimate the remaining parameters of interest, I follow Guimarães et al. (2003) by exploiting the equivalence between this discrete-choice model and a Poisson count-data model. The expected number of entrants from origin o in industry j and location c at time t is then the Poisson intensity

$$\mathbb{E}[m_{ojct}] \;=\; M^E_{ojt} \times \exp\Big(\tfrac{\theta}{\sigma-1}\ln\delta^{-1}_{jc} + \beta' X_{ojct} - \theta \ln \tilde{P}_{jct} + \tfrac{\theta}{\sigma-1}\gamma' Z_{ojct} + \nu_{ojt}\Big). \qquad (17)$$

Since the denominator is common to all commuting zones, it is absorbed in the fixed effect ν_{ojt} defined as:

[3] The $t-5$ subscript used in Table 3 denotes this historical average of lagged values for the agglomeration and export-share regressors; the wage $w_{c,t-5}$ is a strict one-wave lag.

[4] For waves before 2009 I use the 2005 port-activity distances, the earliest the customs data cover.

$$\nu_{ojt} \;=\; -\ln\sum_{c'}\exp\big[\tfrac{\theta}{\sigma-1}\ln\delta_{jc'}^{-1}+\beta' X_{ojc't}-\theta\ln\tilde{P}_{jc'}+\tfrac{\theta}{\sigma-1}\gamma' Z_{ojc't}\big].$$

Domestic firms. Domestic firms have no location choice. For a pool of potential entrepreneurs M^E_{djct}, the number of entrants m_{djct}, new Mexican-owned plants, depends on the probability that an entrepreneur draws a productivity high enough to cover the fixed cost of operating in location c.

The pool is not observed. I proxy it by the national birth rate of industry j at time t, b_{jt}, times the industry's lagged occupied labor in the CZ:

$$M^E_{djct} = b_{jt}\times L^O_{jc,t-5}.$$

The birth rate is the number of new establishments of any ownership in industry j at time t over the average number of establishments in the industry at t and $t-5$. The number of entrants is then

$$m_{djct} = M^E_{djct}\times\mathbb{P}(z_{djc}\geq z^{d,*}_{jc}),$$

where $z^{d,*}_{jc}$ is the domestic productivity cutoff at which profits in location c are zero. From (4) and the revenue expression above,

$$z^{d,*}_{jc} \;=\; \left(\frac{\sigma\, f^d_c}{A_j}\right)^{\frac{1}{\sigma-1}}\tilde{P}_{jc}, \qquad A_j \;\equiv\; Y^d_j\left(P^d_j\right)^{\sigma-1}\left(\tfrac{\sigma}{\sigma-1}\right)^{1-\sigma}\phi_d^{\sigma-1},$$

so the cutoff is proportional to the input-price bundle. Under a constant fixed cost and the rare-event approximation to the Fréchet survival probability, the probability of drawing a productivity above it delivers a Poisson count model (Appendix B):

$$\mathbb{E}[m_{djct}] \;=\; M^E_{djct}\times\exp\Big(\beta' X_{djct}-\theta\,\ln\tilde{P}_{jct}+\nu_{jt}+\zeta_c\Big), \tag{18}$$

where θ is the Fréchet productivity shape parameter. As with foreign firms, the vector X_{djct} includes measures of both origin-specific and industry-specific agglomeration, capturing these effects through the scale parameter T_{djc}; all domestic firms share one origin group, so the domestic origin measure is the domestic stock in the CZ.

Parameter recovery. The model has a single Fréchet shape θ and a single CES elasticity σ, and each is recovered wave by wave from a single moment: the median markup for σ, the mean/median ratio of the productivity distribution for θ. The elasticity of substitution σ is not identified from either entry equation under a constant fixed cost (Appendix B; in

the foreign equation it only scales the trade coefficients and the exit term); it is recovered from firm-level markups (Antràs et al., 2017). Within each census wave I take the median firm-level markup $\hat{\mu}_t = \text{med}_f\,(R_{ft}/\text{VC}_{ft})$, with VC_{ft} the sum of labor, materials and energy costs, over establishments reporting all three inputs and with markups trimmed to $[1, 10]$. I invert it to a wave-specific $\hat{\sigma}_t = \hat{\mu}_t/(\hat{\mu}_t - 1)$; the reported $\hat{\sigma}$ is the average of these values across the 1999–2024 waves, with the cross-wave standard deviation as its dispersion.

The shape θ comes from the size distribution of establishments. Following Hsieh et al. (2013), who recover a Fréchet shape from the coefficient of variation of wages within occupations, I invert a moment ratio that does not depend on the scale of the distribution. I use the ratio of the mean to the median, which only requires $\theta > 1$:

$$\frac{\mathbb{E}[z]}{\text{med}(z)} \;=\; \Gamma\big(1 - 1/\theta\big)\,(\ln 2)^{1/\theta} \;\equiv\; g(\theta), \qquad \theta > 1. \tag{19}$$

The function g is strictly decreasing, from infinity at $\theta = 1$ to one as $\theta \to \infty$: the mean sits above the median because the most productive draws pull it up, and the less dispersed the draws, the smaller the gap. For this reason, each observed mean/median ratio maps into a single value of θ, and I do not need to know T_{ojc}, which differs across cells, to recover it.

Productivity itself is not observed in the census. The model, however, ties it to employment. Within a cell (o, j, c, t) every plant faces the same input prices, demand and markup, so by (2) and CES demand a non-exporter's revenue is proportional to $(\phi_o z_i)^{\sigma-1}$, and by the labor share of Appendix A so is its employment. Hence

$$L_i \;\propto\; (\phi_o z_i)^{\sigma-1}, \qquad \hat{z}_i \;\equiv\; L_i^{1/(\hat{\sigma}-1)} \;\propto\; \phi_o\, z_i.$$

The $(\sigma - 1)$-th root of employment is the productivity draw times a constant common to the cell's plants, and a common constant multiplies the mean and the median alike, so it drops out of (19). An exporter's revenue carries one more factor, common to the cell's exporters but not to its other plants; it is one of the sources of extra dispersion in the lower-bound remark below. Taking the root is important. Employment itself is Fréchet with shape $\theta/(\sigma - 1)$, which is close to one, so its variance does not exist and its sample mean is dominated by the largest plants. The transformed $\hat{z}$, in contrast, has shape θ.

I compute the estimator separately for each wave, in four steps. (i) For every establishment with positive paid employment, compute $\hat{z}_i = L_i^{1/(\hat{\sigma}-1)}$ with the pooled $\hat{\sigma}$, the same in every wave; paid employment is used because it is not winsorized. (ii) Keep only the origin $\times$ industry $\times$ CZ cells with at least 50 establishments and, within each of them, divide $\hat{z}_i$ by the cell median. After this step the median plant of each cell equals one, and the scale of the cell is gone. (iii) Average these ratios over all plants in the kept cells to get $\bar{\rho}_t$. Because θ is

the same in every cell, $\bar{\rho}_t$ is an estimate of the left side of (19). (iv) Solve

$$g(\hat{\theta}_t) \;=\; \bar{\rho}_t \tag{20}$$

for $\hat{\theta}_t$ numerically, since g has no closed-form inverse, and average across waves, with the cross-wave standard deviation as the dispersion, as for σ. I only keep cells with 50 or more plants because the sample median of a small cell is noisy. In simulated cells of 20 plants this noise biases $\hat{\theta}$ upward by about 3%, and with 50 plants the bias is about 1%.

I make two remarks on this estimator. First, the same inversion works for foreign and Mexican-owned cells, because selection does not change the shape of the distribution. The draws of the foreign plants that chose a given CZ are again Fréchet with shape θ, and plants that are only observed above an operating cutoff follow a Pareto distribution in the tail with the same θ. For θ between 2.5 and 6, the mean/median ratio of this Pareto is within 2% of $g(\theta)$. Second, any dispersion within a cell other than the draws, such as differences in ϕ_o across firms of an origin, export sales, or measurement error, raises $\bar{\rho}_t$, so $\hat{\theta}$ is a lower bound.

Estimation. I estimate both (17) and (18) by Poisson pseudo-maximum likelihood (PPML), which fits the exponential mean of a count model without assuming a Poisson distribution and absorbs the high-dimensional fixed effects exactly (Correia et al., 2020, Santos Silva and Tenreyro, 2006), with standard errors clustered at the commuting-zone level. PPML drops singleton observations, observations separated by the fixed effects (fixed-effect groups with no entrants) and cells whose offset is undefined, so the estimation samples (6,139 foreign and 41,780 domestic cells) are smaller than the cell counts of Table 2. The shape θ is not estimated from the entry equations. It is recovered from the size distribution of establishments as described above and imposed on both. The input-price term $-\hat{\theta}\ln\tilde{P}_{jct}$ enters each equation as part of the offset, and so does the exit term $\frac{\hat{\theta}}{\hat{\sigma}-1}\ln\delta_{jc}^{-1}$ in the foreign equation: the exit rate scales the value of a location as $1/\delta_{jc}$, which the Fréchet draws raise to $\theta/(\sigma-1)$ in the choice probability, as they do the export factor. The PPMLs therefore estimate only the agglomeration loadings and, in the foreign equation, the trade terms.

Foreign equation. The origin×industry×year fixed effect absorbs ν_{ojt} exactly. Under the input-price assumptions detailed in Appendix A, the spatial variation in $\ln\tilde{P}_{jct}$ collapses to $\beta_{Lj,t}\ln w_{c,t-5}$, with $w_{c,t-5}$ the CZ wage bill over paid employment across all census sectors in the previous wave, lagged so that the offset term is predetermined like the agglomeration measures; the industry-year component of the input-price bundle in (27) is absorbed by the fixed effect. The coefficients on X_{ojct} identify the loadings of T_{ojct} on origin- and industry-specific agglomeration; the coefficients on Z_{ojct} identify the trade-cost components scaled by

$\theta/(\sigma-1)$. The exit term $\ln \delta_{jc}^{-1}$ is constructed from the census panel as the log-inverse of the average exit rate observed in each industry-CZ cell across waves, a fixed cell characteristic as in the model; unlike the lagged wage it is not predetermined, since it averages over all waves.

Domestic equation. Fixed effects are simpler: an industry-year fixed effect ν_{jt} absorbs time-varying industry-level unobservables, including industry expenditure Y_j^d and the industry price index P_j^d, and a CZ fixed effect ζ_c absorbs time-invariant location-level unobservables, including the fixed-cost requirement f_c^d (Appendix B). With a CZ rather than a CZ-year fixed effect, the domestic same-origin stock varies within CZs over time, so the domestic equation can in principle identify both agglomeration loadings, although the origin loading turns out imprecise (Section 4.3).

4.3 Results

Table 3 reports the two firm-entry equations side by side, each under three agglomeration specifications: the same-origin stock alone, the same-industry stock alone, and both together. The foreign columns report the PPML estimates of (17); the domestic columns report (18). Both impose $\hat{\theta} = 2.789$ (Table 4) through the offset, and the foreign columns also carry the exit term there. Since the model has both agglomeration channels, the counterfactual uses the specifications in columns (3) and (6). I still report the single-channel columns, because comparing them shows how little independent variation the origin channel has.

What the specifications show. Once the industry channel is included, the origin channel loses most of its loading. On its own the origin loading looks large and precise (0.229 in column (1)). Adding the same-industry stock, however, cuts it by a factor of almost four, to 0.062 in column (3), whereas the industry loading changes much less between columns (2) and (3) (0.435 to 0.369). The shared export core of Section 2.3 is the reason: because foreign firms of every origin concentrate in the same few industries, with a handful of plants in each industry and CZ, a cell's same-origin stock is mostly its same-industry stock, so an origin-only specification attributes the industry force to the origin label. The origin loading is therefore identified only from the origin variation that is left within industry clusters. The main agglomeration force is the industry loading (0.369), and the origin channel survives only as a small but precisely signed 0.062.

Only one term on the export margin is precisely estimated. The interaction of the lagged export-revenue share with the distance to the nearest seaport is negative in all three foreign columns (-0.284 in column (3), significant at the 5% level), so among export-oriented cells there is less entry farther away from a port. The coefficients on the standalone share and

Table 3: Foreign and Domestic Firm Entry: Agglomeration Specifications (wL)

	Foreign m_{ojct}			Domestic m_{djct}		
	Origin	Industry	Both	Origin	Industry	Both
	(1)	(2)	(3)	(4)	(5)	(6)
Origin Agglomeration ($\ln(1+wL^{o,c}_{t-5})$)	0.229***	—	0.062***	−0.239**	—	0.053
	(0.050)		(0.016)	(0.094)		(0.069)
Industry Agglomeration ($\ln(1+wL^{j,c}_{t-5})$)	—	0.435***	0.369***	—	−0.302***	−0.303***
		(0.046)	(0.039)		(0.019)	(0.019)
$\omega^{x}_{ojc,t-5}$	3.170	1.928	1.805	—	—	—
	(2.096)	(1.713)	(1.747)			
$\omega^{x}_{ojc,t-5} \times \ln d^{\text{sea}}$	−0.363***	−0.271**	−0.284**	—	—	—
	(0.136)	(0.131)	(0.126)			
$\omega^{x}_{ojc,t-5} \times \ln d^{\text{border}}$	−0.042	−0.063	−0.047	—	—	—
	(0.127)	(0.103)	(0.106)			
$\omega^{x}_{ojc,t-5} \times \ln d^{\text{air}}$	−0.222	−0.088	−0.088	—	—	—
	(0.263)	(0.219)	(0.223)			
Constant	−15.262***	−17.729***	−17.649***	5.037***	5.182***	4.452***
	(0.550)	(0.518)	(0.533)	(1.305)	(0.214)	(1.039)
Origin × Industry × Year FE	X	X	X	—	—	—
Industry × Year FE	—	—	—	X	X	X
CZ FE	—	—	—	X	X	X
Offset: $\ln M^{E}$	X	X	X	X	X	X
Offset: $-\hat{\theta}\ln\tilde{P}_{jct}$	X	X	X	X	X	X
Offset: $\frac{\hat{\theta}}{\hat{\sigma}-1}\ln\delta^{-1}_{jc}$	X	X	X	—	—	—
Observations	6,139	6,139	6,139	41,780	41,780	41,780
Pseudo R^2	0.124	0.197	0.202	0.858	0.892	0.892
Counterfactual specification	—	—	**X**	—	—	**X**

Note: PPML on entrant counts, 1999–2024. $\hat{\theta} = 2.789$ (Table 4); the offset carries the spatial part $\beta_{Lj,t}\ln w_{c,t-5}$ of $\ln\tilde{P}_{jct}$ and, in the foreign columns, the exit term $(\hat{\theta}/(\hat{\sigma}-1))\ln\delta^{-1}_{jc}$. SE clustered by CZ. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

on the border and airport interactions are not significant. A possible explanation for the missing border term is that the infrastructure at the border works through agglomeration itself. The first *maquiladoras* settled close to the US border, and the industry agglomeration they built is what now attracts new entrants. If that is the case, there is little left for the distance interaction to capture.

In the domestic equation, the industry loading is *negative* (−0.303) and precisely estimated. It is almost the same when I drop the origin stock (−0.302), and it also survives other fixed-effect designs, which I do not report. To read it, keep in mind how the pool is built. The pool of potential entrants is proportional to the industry's lagged labor in the CZ, so the loading measures how entry grows relative to the size of that pool. Once the pool is added back, domestic entry rises with local industry size with an elasticity of about 0.70, which is less than one for one. In the counterfactual I keep the pool at its baseline value (Appendix D). The domestic origin loading is weakly identified: under the CZ fixed effect its only variation is within-CZ growth of the domestic stock, which moves with overall CZ growth, so it is −0.239 alone and 0.053, insignificant, next to the industry stock.

Structural parameters. Table 4 reports the wave-by-wave inversions behind $\hat{\sigma}$ and $\hat{\theta}$.

Table 4: Structural Parameters by Census Wave

	σ from markups		θ from the size distribution	
Wave	$\hat{\sigma}_t$	Establishments	$\hat{\theta}_t$	Establishments
1999	3.329	137,372	2.577	113,451
2004	3.380	130,285	2.677	104,177
2009	3.646	153,258	2.817	135,163
2014	3.196	133,148	2.747	123,285
2019	3.130	193,009	2.921	172,506
2024	3.556	225,357	2.997	197,999
Mean	3.373	972,429	2.789	846,581
SD across waves	(0.200)		(0.156)	

Note: $\hat{\sigma}_t$ inverts the wave's median markup over labor, materials and energy; $\hat{\theta}_t$ inverts the mean of $\hat{z}_i$ over its cell median in cells of 50 or more establishments. Establishment counts are summed in the Mean row.

The recovered parameters are $\hat{\sigma} = 3.373$ (cross-wave SD 0.200) and $\hat{\theta} = 2.789$ (SD 0.156). The integrability condition $\theta > \sigma - 1$ holds, and it holds wave by wave: $\hat{\theta}_t$ exceeds $\hat{\sigma}_t - 1$ in all six waves. The implied tail index of establishment size, $\hat{\theta}/(\hat{\sigma} - 1) = 1.18$, is close to the Zipf value of one commonly found for firm sizes. $\hat{\theta}$ sits between the firm-level Fréchet shape of Antràs et al. (2017) ($\hat{\theta} \approx 1.8$, estimated on sourcing rather than location choice) and the aggregate trade elasticity of about 5 recovered from country-level gravity (Head and Mayer,

2014), and $\hat{\sigma}$ lies within the range of median elasticities of substitution among imported varieties in Broda and Weinstein (2006) (2.7 to 3.6). The shape rises over the period, from 2.58 in 1999 to 3.00 in 2024 with one dip in 2014: productivity draws within cells have become less dispersed.

5 Counterfactual Analysis

The estimates say foreign exporters sit in several distinct corridors and industry agglomeration, and origin agglomeration to a lesser extent, keeps them there. A tariff on exports takes the same proportion of export revenue from every exporter that pays it. Any spatial pattern in the loss must therefore come from two places, namely where the exporters are and how each local labor market absorbs the shock, and in this section I measure both across commuting zones. Section 5.1 presents the two scenarios I consider, a tariff on all exporters and a tariff on foreign-owned exporters only. The second scenario tells me how much of the damage is due to foreign exporters alone. For both, I report the mean welfare change across CZs and how it splits into an income and a price channel. In Section 5.2 I derive the responses of firm mass, productivity scale and the input price in each CZ, and I show how the dispersed footprint of foreign firms from Section 2 becomes a dispersed welfare loss through export exposure and the wage adjustment that follows. Section 5.3 maps these losses, which concentrate in the foreign-firm corridors of Section 2. Finally, Section 5.4 breaks the welfare change down step by step into direct, wage, mass and agglomeration terms, so that the agglomeration feedback can be seen on its own.

5.1 The Aggregate Blow

Setup. The central case imposes an ad valorem tariff of $\kappa = 0.25$ on the exports of every tradable industry in the sample. This is the rate the United States applied from March 2025 to Mexican goods that did not qualify under the United States-Mexico-Canada Agreement (USMCA), under the International Emergency Economic Powers Act (IEEPA); Mexico was exempt from the April 2025 reciprocal tariffs and stayed under this order until the Supreme Court held in February 2026 that IEEPA does not authorize tariffs. Goods that qualified under USMCA did not pay this tariff, so a single rate on all exports overstates what was actually applied, and I read the results as an upper bound. My baseline equilibrium is the 2024 census cross-section, and I look at two cases. Under the *all-exporter* scenario, every exporting firm pays the tariff, whether it is foreign or domestic. Under the *foreign-only* scenario, only foreign-owned exporters do. The tariff raises the export-market iceberg trade

cost to $\tau_{jc}^{x,cf} = \tau_{jc}^{x}(1+\kappa)$, which enters the numerator of the foreign choice probability in (16). Since it is modelled as an iceberg wedge, it also lowers the export revenue of every taxed exporter to $r^{x,\text{tariff}} = r^{x}\,(1+\kappa)^{1-\sigma}$. The total mass of foreign entrants of each origin-industry pair stays at its baseline level. What changes is where they go, through the choice probability, and how many of them clear the cutoff, which depends on profits. On the domestic side, the mass of Mexican-owned firms moves with the productivity cutoff through the domestic Poisson intensity, while the pool of potential entrants stays at baseline. The new equilibrium is the exact-hat fixed point described in Appendix E.

I also repeat both scenarios for $\kappa \in \{0.10, 0.25, 0.50\}$, to see how much the rate matters. The low case is the 2026 rate on Mexican goods that do not qualify under USMCA: a 10% surcharge under Section 122 of the Trade Act of 1974 from February to July 2026, followed by a 10% duty under Section 301 since July 2026. For the high case I use $\kappa = 0.50$, which is a prohibitive upper bound. The full welfare decomposition below is for the central case, and the sensitivity panel at the bottom of Table 5 gives aggregate welfare and the income channel at each of the three rates.

Decomposition of the welfare effect. Because domestic trade is costless, P is common across CZs, so all spatial variation in $d\ln W_c$ under the tariff comes from the income channel; the price channel shifts the level of welfare uniformly, though it aggregates responses from every CZ. Totally differentiating (14) using the price index definitions (12) and (13) yields

$$d\ln W_c = \underbrace{d\ln(w_c L_c + \Pi_c^d)}_{\text{Income channel}} + \underbrace{\chi \sum_j \frac{\alpha_j}{\sigma-1}\, d\ln\left(\sum_c M_{jc}\right)}_{\text{Variety}} + \underbrace{\chi \sum_j \frac{\alpha_j}{\theta}\, d\ln\left(\sum_c T_{jc}\,\tilde{P}_{jc}^{-\theta}\right)}_{\text{Productivity \& input price}}. \quad (21)$$

The two price sub-channels aggregate to $-\chi\, d\ln P^M$. Both follow the standard CES logic. The industry price index falls with more firms (variety), with more productive firms (productivity) and with cheaper inputs (input price), and welfare rises whenever one of these lowers P^M. Since I hold the non-tradable price at its baseline value, $d\ln P_S = 0$ and this term drops out. The masses move as in Section 5.2. Mexican-owned plants enter and exit through (22), and foreign plants relocate and exit through (24), while the national pool of foreign entrants stays fixed. The decomposition is CES price-index accounting applied to this index: the variety term is the love-of-variety term of Krugman (1980) with elasticity $1/(\sigma-1)$, and the productivity term carries the Fréchet elasticity $1/\theta$ of Eaton and Kortum (2002).

Inside the second price term, T and $\tilde{P}$ are entangled through the sum $\sum_c T_{jc}\tilde{P}_{jc}^{-\theta}$: at the industry-aggregate level, this does not separate into a term involving only T_{jc} and a term involving only $\tilde{P}_{jc}$. Cell-by-cell, however, the log differential is separable, $d\ln(T_{jc}\tilde{P}_{jc}^{-\theta}) =$

$d \ln T_{jc} - \theta \, d \ln \tilde{P}_{jc}$, so aggregating across CZs with the productive-capacity weights $\omega_{jc} \equiv T_{jc}\tilde{P}_{jc}^{-\theta} / \sum_{c'} T_{jc'}\tilde{P}_{jc'}^{-\theta}$ yields

$$d \ln\Big(\textstyle\sum_c T_{jc}\tilde{P}_{jc}^{-\theta} \Big) \;=\; \textstyle\sum_c \omega_{jc} \, d \ln T_{jc} \;-\; \theta \sum_c \omega_{jc} \, d \ln \tilde{P}_{jc}.$$

The levels of T_{jc} are absorbed by the fixed effects of the entry equations, so I measure ω_{jc} by each CZ's share of the industry's baseline domestic sales.

Headline numbers. Under a 25% tariff on all Mexican tradable exports, the income-weighted average commuting zone loses 0.042 log points of income (cross-CZ standard deviation 0.033), whereas restricting the tariff to foreign-owned exporters cuts the income loss to 0.018 (SD 0.017). Adding the aggregate price channel, 0.010 and 0.005 respectively, welfare falls by 0.051 in the all-exporter case and by 0.023 in the foreign-only case. The price channel is the same everywhere, so the differences between CZs come only from income. Without income weights the mean welfare change is -0.027, much smaller than the weighted -0.051. This is because the CZs that export the most are also the ones with high incomes, and they carry most of the loss. Table 5 reports these totals in its last rows. The rows above them decompose the change into the income and price channels, again weighting CZs by income. In the sensitivity block, I show the same aggregates for $\kappa \in \{0.10, 0.50\}$. The income loss rises less than proportionally with κ (-0.020, -0.042, -0.064), while the price term grows with it (-0.005, -0.010, -0.015).

Income channel: wages carry the loss. In the all-exporter case, -0.038 of the 0.042 income loss is the wage sub-channel (Table 5). Domestic profits account for only -0.003. Equation (11) explains why the wage moves less than tradable revenue. A shock that only reaches tradables moves the wage by λ_c, the tradable share of local labor payments, and this share averages 0.36 with wage-bill weights (Table 6; 0.32 unweighted). At the average λ_c, then, the wage takes up about a third of the tradable revenue shock. Profits fall less. Mexican-owned exporters lose export revenue (-0.011 with wages and masses frozen, Table 7), but new Mexican-owned plants enter through (22) where the wage falls and where the industry stock thins, and this entry recovers three-quarters of the loss. In the foreign-only case income still falls by 0.018, more than two-fifths of the all-exporter loss, and it does so through the wage. Foreign firms produce three-fifths of exports, and the local wage reacts to a smaller tradable payroll regardless of who owns the plants. Domestic profits even rise in this case ($+0.007$). The profits of the taxed foreign plants do not go to Mexican households, and Mexican-owned cells pay no tariff, so their revenue only changes through the domestic mass response (22) (Appendix D), which the lower wage pushes up.

Table 5: CZ-Level Welfare Decomposition, Tariff on Mexican Manufacturing Exports

	All-exporter		Foreign-only	
Channel / component	mean (sd)	median	mean (sd)	median
Central case: $\kappa = 0.25$ (2025 IEEPA rate)				
Income channel				
Total ($d \ln Y_c$)	−0.042 (0.033)	−0.034	−0.018 (0.017)	−0.008
Wage sub-channel	−0.038 (0.030)	−0.026	−0.027 (0.022)	−0.014
Profit sub-channel	−0.003 (0.011)	−0.003	0.007 (0.007)	0.006
Price channel, aggregate (CZ-invariant)				
Total ($-\chi\, d \ln P^M$)	−0.010	−0.010	−0.005	−0.005
Variety	−0.016	−0.016	−0.009	−0.009
Productivity	−0.001	−0.001	−0.001	−0.001
Input price	0.007	0.007	0.005	0.005
Price channel, CZ-local				
Total	−0.005 (0.004)	−0.007	−0.002 (0.003)	−0.003
Variety	−0.011 (0.004)	−0.010	−0.005 (0.003)	−0.005
Productivity	0.000 (0.001)	0.000	−0.000 (0.001)	0.000
Input price	0.005 (0.004)	0.004	0.004 (0.003)	0.002
Total welfare change				
$d \ln W_c$ (income-weighted)	−0.051 (0.033)	−0.044	−0.023 (0.017)	−0.012
$d \ln W_c$ (unweighted)	−0.027 (0.033)	−0.010	−0.011 (0.017)	−0.005
Sensitivity: aggregate welfare and income channel across κ (income-weighted)				
$\kappa = 0.10$ (2026 rate on non-USMCA goods)				
$d \ln W_c$	−0.025 (0.016)	−0.022	−0.012 (0.008)	−0.007
Income channel	−0.020 (0.016)	−0.017	−0.010 (0.008)	−0.005
$\kappa = 0.50$ (prohibitive upper bound)				
$d \ln W_c$	−0.079 (0.052)	−0.067	−0.033 (0.024)	−0.018
Income channel	−0.064 (0.052)	−0.051	−0.026 (0.024)	−0.011

Note: Log points; negative = loss. Income-weighted means and medians, unweighted standard deviations across CZs, trimmed as in Appendix E.2 (Tables 6 and 7 untrimmed). CZ-local rows are a diagnostic and do not enter $d \ln W_c$; rows may not sum because of rounding.

Price channel: fewer plants, cheaper inputs. The aggregate price channel is −0.010 under the all-exporter tariff and −0.005 under the foreign-only tariff, because the tariff raises the tradable price index. The variety term (−0.016 and −0.009) is there because plants that lose their export revenue stop clearing the cutoff and close. Productivity barely moves (−0.001 in both scenarios). With domestic-sales weights, the taxed exporters whose wage-bill stocks erode have little weight in the price index. At the same time, Mexican-owned cells gain scale where the industry stock thins, because their industry loading is negative (Mechanism 2 of Section 5.2). The input-price term goes in the opposite direction (+0.007 and +0.005). Lower wages make local inputs cheaper, which lowers the price index and offsets two-fifths and half of the variety loss, respectively. The CZ-local rows of the table show the same pattern, with modest dispersion across CZs.

5.2 Why the Reach Is So Broad

Foreign firms make up less than 1% of Mexican manufacturing establishments, and yet a shock to them lowers welfare in most of the country's high-income regions at the same time. Two features of the model are behind this. One is how exposed the corridors are at baseline. Each of the origin corridors of Section 2 has large exporters, so a single export tariff lowers labor demand, and with it the market-clearing wage, in all of them at the same time. The other is that the three CZ-level responses I derive below, in firm mass M_c, in productivity scale T_c and in the input-price bundle $\tilde{P}_c$, end up in the same place: the industry price index P_j^d of (12), which adds up firm masses, productivity scales and input bundles cell by cell. Since P_j^d is the same in every CZ (Section 5.1), these responses mostly enter as a price-channel term that is common across space, and add little to the geographic incidence of the loss. In what follows I state the three response equations, plot their CZ-local welfare counterparts in Figure 3, and end with a decomposition by exposure (Table 6) that links the geographic footprint of foreign firms to the geographic footprint of the welfare loss.

The response equations below are exact-hat ratios $\widehat{x} \equiv x'/x$ of the estimated equations (Caliendo and Parro, 2015, Dekle et al., 2008), derived in Appendix D. These ratios only require baseline data, namely revenues, choice probabilities, wages and labor stocks, and not the levels of T_{ojc} or $\tilde{P}_{jc}$. Every response is therefore relative to the observed baseline. For instance, $\widehat{M}_c > 1$ or $\widehat{T}_c > 1$ means that the counterfactual pushes the object above its baseline value, and not that the object is high in absolute terms.

Mechanism 1: firm mass responds to exposure. Total firm mass at CZ c splits into a domestic component and a foreign component. Each has a hat equation obtained from its Section 4 entry equation by this ratio operation. In both components the tariff enters

through the export-return factor $\widehat{\varepsilon}_{ojc} \equiv \left[(1-\omega^x_{ojc}) + \omega^x_{ojc}(1+\kappa)^{1-\sigma}\right]^{\theta/(\sigma-1)}$, where ω^x_{ojc} is the cell's baseline export share of revenue. This ratio of counterfactual to baseline export returns summarizes how the tariff affects revenue. Since the export revenue of every taxed exporter is multiplied by $(1+\kappa)^{1-\sigma}$, the value of a cell falls more the more it exported at baseline.

Domestic firm mass. Applying this to the domestic Poisson intensity (18), with the export term that the cutoff on total profit adds, gives

$$\widehat{M^d_{jc}} \;=\; \widehat{T_{djc}}\;\widehat{\tilde{P}}_{jc}^{\,-\theta}\;\widehat{\varepsilon}_{djc}, \tag{22}$$

where $\widehat{T_{djc}}$ is the change in the cell's Fréchet scale, given by the agglomeration equation (25) below with the domestic loadings, and the entrant pool $M^E_{djct} = b_{jt}L^O_{jc,t-5}$ is held at its baseline value. It is through the export-return factor $\widehat{\varepsilon}_{djc}$ that the tariff shuts down plants, because a Mexican-owned exporter that loses part of its export revenue may no longer make it past the cutoff. For the exponent $-\theta$ I use the common Fréchet shape $\hat{\theta} = 2.789$ of Table 4. Since this exponent is negative, a lower wage attracts domestic entrants.

Foreign firm mass. On the foreign side, the number of entrants is $m_{ojct} = M^E_{ojt}\,\mathbb{P}(c \mid ojt)$. Here M^E_{ojt} is the exogenous pool of potential entrants and $\mathbb{P}(c \mid ojt)$ the share of that pool that picks CZ c. In the counterfactual I keep M^E_{ojt} at its baseline value (Appendix E.2, Step 2), so the entrant count can only change through the choice probability, and $\widehat{m}_{ojct} = \widehat{\mathbb{P}}(c \mid ojt)$. Applying the ratio operation to (16) yields

$$\widehat{\mathbb{P}}(c \mid ojt) \;=\; \frac{\widehat{T_{ojc}}\;\widehat{\tilde{P}}_{jc}^{\,-\theta}\;\widehat{\varepsilon}_{ojc}}{\sum_{c'}\mathbb{P}(c' \mid ojt)\,\widehat{T_{ojc'}}\;\widehat{\tilde{P}}_{jc'}^{\,-\theta}\;\widehat{\varepsilon}_{ojc'}}, \tag{23}$$

where $\mathbb{P}(c' \mid ojt)$ are the baseline choice probabilities, so the denominator is the baseline-share-weighted mean of the numerator hats across CZs. Dividing by the mean makes sure that the probabilities still add up to one after the shock, $\sum_c \mathbb{P}(c \mid ojt) = 1$. CZs whose numerator hat is above the mean gain probability, and those below it lose. The tariff shows up in this expression through $\widehat{\varepsilon}_{ojc}$: the more CZ c exported at baseline, the more its value to an origin-o industry-j firm falls. In general equilibrium the choice probability also moves through $\widehat{T}$ and $\widehat{\tilde{P}}$. When firm masses relocate, the agglomeration measures change (Mechanism 2 below) and T moves with them. When labor demand shifts across CZs, the market-clearing wage changes and passes through to $\tilde{P}$ (Mechanism 3 below). All three of these feed back into $\widehat{\mathbb{P}}$ until the fixed point converges.

Foreign exit. Foreign plants also have to cover their fixed costs in the CZ they chose, so the tariff closes some of them as well, namely those whose draws no longer clear the

cutoff. The plants in c hold the draws that won the comparison, whose value has the same distribution whichever CZ won, so under the rare-event approximation used for domestic entry (Appendix B) the share of them that clears the cutoff moves with the pair's total attractiveness, the denominator of (23), and the foreign operating mass is

$$\widehat{M^{f}_{ojc}} \;=\; \widehat{\mathbb{P}}(c \mid ojt) \sum_{c'} \mathbb{P}(c' \mid ojt)\, \widehat{T_{ojc'}}\, \widehat{\tilde{P}}^{-\theta}_{jc'}\, \widehat{\varepsilon}_{ojc'} \;=\; \widehat{T_{ojc}}\, \widehat{\tilde{P}}^{-\theta}_{jc}\, \widehat{\varepsilon}_{ojc}. \tag{24}$$

Relocation moves entrants away from the CZs the tariff hits hardest, and exit lowers the pair's operating mass by that same factor in every CZ.

Summing the two. The two sub-channels aggregate to $d\ln M_c$ with baseline revenue weights: within each industry of the CZ, the Mexican-owned and foreign mass hats are averaged by baseline cell revenue, and the industries by their expenditure shares α_j. Foreign plants are few but large, so the foreign hat weighs far more in $d\ln M_c$ than the foreign share of plants.

Mechanism 2: productivity reshuffles across CZs. Applying the same relative-changes logic of Caliendo and Parro (2015) to the estimated agglomeration equation gives the productivity response as a function of the mass hats from Mechanism 1, again without requiring the baseline level of T_{ojc}. The Fréchet scale is parameterized in the estimation as $\ln T_{ojc} = \beta_{og}\, \ln(1 + wL^{o,c}) + \beta_j\, \ln(1 + wL^{j,c})$, with the stocks of Section 4, the loadings of the foreign PPML for foreign cells and those of the domestic PPML for Mexican cells. Under Cobb-Douglas cost minimization a cell's wage bill is a fixed share of its revenue, so the wage bill of cell (o, j, c) moves with its revenue hat, which is the mass hat itself, $\widehat{R}_{ojc} = \widehat{M_{ojc}}$: the average revenue of the plants that clear the cutoff is a fixed multiple of the fixed cost (Appendix D), so the tariff reaches cell revenue, and the stocks, through the plants that enter, exit or relocate. Each stock moves by the baseline-weighted change of the cells it contains,

$$\widehat{wL^{o,c}} \;\equiv\; \frac{\sum_j wL_{ojc}\, \widehat{R}_{ojc}}{\sum_j wL_{ojc}}, \qquad \widehat{wL^{j,c}} \;\equiv\; \frac{\sum_o wL_{ojc}\, \widehat{R}_{ojc}}{\sum_o wL_{ojc}},$$

where wL_{ojc} is the observed baseline wage bill of cell (o, j, c) and $\widehat{M_{ojc}}$ combines the domestic hat (22) and the foreign hat (24). Feeding these into the estimated agglomeration equation:

$$\ln \widehat{T_{ojc}} \;=\; \beta_{og}\, \ln\!\left(\widehat{wL^{o,c}}\right) \;+\; \beta_j\, \ln\!\left(\widehat{wL^{j,c}}\right), \tag{25}$$

with the foreign loadings for foreign cells and the domestic loadings for Mexican cells; for Mexican cells the origin stock is the domestic stock in the CZ. Because the domestic industry

loading is negative, $\widehat{T_{djc}}$ by itself rises where the industry stock falls, and Section 4.3 gives the net elasticity once the pool is included. Under the tariff, $\widehat{M_{ojc}}$ falls in exporter-heavy CZs and moves little in interior CZs, and $\widehat{wL^{o,c}}$ and $\widehat{wL^{j,c}}$ follow the same pattern. The scale of foreign cells falls where their stocks shrink. Mexican-owned cells have a negative industry loading, so their scale goes up where the industry stock gets thinner. With domestic-sales weights these two effects almost offset each other. As a result, $d\ln T_c$ is within 0.01 of zero in every exposure group (Table 6), and the productivity panel of Figure 3 is centered on zero.[5]

Mechanism 3: the wage carries the shock into the input bundle. By (27) only the wage varies across CZs, so the input bundle moves with it, and labor-market clearing (11) at fixed L_c and $\widehat{R}_{Nc} = 1$ pins down the wage hat:

$$\widehat{w}_c \;=\; \lambda_c\,\widehat{R}_{\mathcal{T}c} + (1-\lambda_c), \qquad \widehat{\tilde{P}}_{jc} \;=\; \widehat{w}_c^{\,\beta_{Lj}}, \tag{26}$$

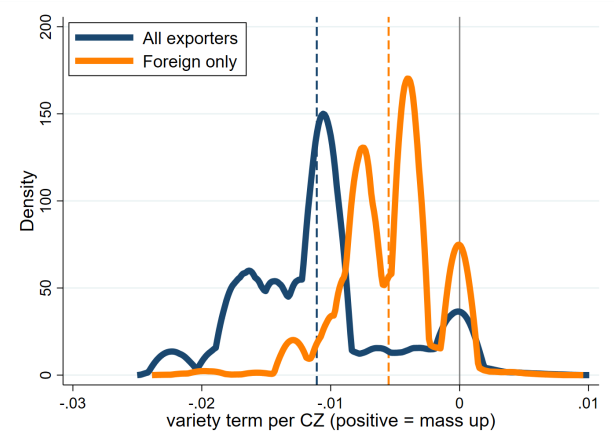
where λ_c is the tradable share of the CZ wage bill and $\widehat{R}_{\mathcal{T}c}$ is the wage-bill-weighted mean of the tradable revenue hats (Appendix E.2, Step 5). In words, the wage changes by the tradable revenue shock scaled by λ_c, and the input bundle passes this change through at the labor cost share.

Visualization of the three responses. In Figure 3 I plot the welfare counterparts of the three responses in (22)–(25) and (26). These are the CZ-local variety, productivity and input-price terms of Table 5, which weight the change in each CZ's own firm mass, productivity and input bundle by its price-index elasticity. A positive x means that the underlying object went up. Firm mass falls where foreign exposure is highest, with an income-weighted mean $d\ln M_c$ of -0.073 under the all-exporter tariff and -0.037 under the foreign-only tariff (Table 6). Exporters that lose revenue close, and the lower wage draws in Mexican-owned entrants that offset part of the fall. As Mechanism 2 explains, the productivity panel is centered on zero. The input-price panel is negative, because by (26) local wages fall by the tradable share λ_c of the tradable revenue shock.

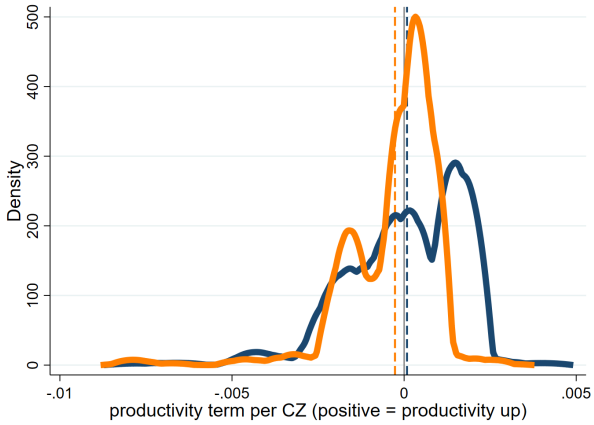
An exposure-based decomposition. Define the baseline foreign revenue share $f_c \equiv R_c^f/(R_c^d + R_c^f)$ and partition CZs into three exposure groups: *no foreign* ($f_c = 0$), *some foreign* ($0 < f_c \leq 0.20$) and *heavy foreign* ($f_c > 0.20$). Table 6 reports income-weighted group means of the two responses derived in Mechanisms 1–2 ($d\ln M_c$, $d\ln T_c$) and of the welfare shift $d\ln W_c$. Firm mass falls more in the more exposed groups, while $d\ln T_c$ stays

[5]The regressor is an expanding mean of past stocks, so applying its loading to the counterfactual change in the stock is a long-run reading; Appendix D states the $\ln(1+wL)$ approximation.

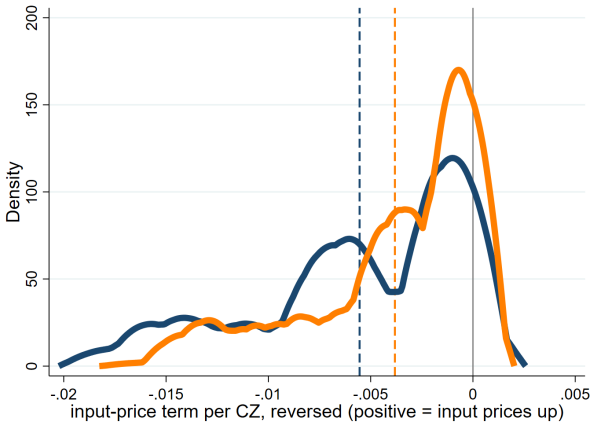
Figure 3: CZ Distribution of the Three Response Variables Through Prices



Mass (variety term)



Productivity (productivity term)



Input price (input-price term, sign reversed)

Note: CZ-local price sub-channels of Table 5, in log points of welfare; the input-price term is shown with its sign reversed, so positive means the object rose. Income-weighted densities, $\kappa = 0.25$, trimmed at the 0.5th and 99.5th percentiles. Dashed vertical lines mark the means.

close to zero in all of them. The welfare loss is concentrated in the heavy-foreign group, which loses 0.079 under the all-exporter tariff, about three times as much as the no- and some-foreign groups, and accounts on its own for three-quarters of the aggregate loss. Under the foreign-only tariff the groups are ranked in the same order and the losses are about half as large, with the two low-exposure groups losing only a little more than the common price term.

Table 6: Exposure Decomposition of the Mean CZ-Level Response

Group	f_c range	N	Income Share	λ_c	$d\ln M_c$	$d\ln T_c$	$d\ln W_c$
All-exporter tariff							
No foreign	$f_c = 0$	483	0.043	0.230	−0.002	−0.000	−0.021
Some foreign	$0 < f_c \leq 0.20$	80	0.455	0.210	−0.060	0.008	−0.026
Heavy foreign	$f_c > 0.20$	175	0.502	0.513	−0.091	−0.007	−0.079
Aggregate		738	1.000	0.363	−0.073	0.000	−0.052
Foreign-only tariff							
No foreign	$f_c = 0$	483	0.043	0.230	0.000	0.000	−0.005
Some foreign	$0 < f_c \leq 0.20$	80	0.455	0.210	−0.025	0.003	−0.008
Heavy foreign	$f_c > 0.20$	175	0.502	0.513	−0.050	−0.007	−0.042
Aggregate		738	1.000	0.363	−0.037	−0.002	−0.025

Note: Income-weighted group means, $\kappa = 0.25$; weighted by income share they sum to the Aggregate row. f_c: baseline foreign share of CZ tradable revenue. λ_c: baseline tradable share of the CZ wage bill, group means weighted by the wage bill.

5.3 Where the Loss Lands

Given the exposure gradient of Table 6, losses should follow the baseline foreign footprint of Section 2, scaled by the tradable share of each local wage bill. Figure 4 maps the change in welfare and income by CZ, and the losses do concentrate where the mechanism says they should, for instance in Ciudad Juárez, Tijuana, Nuevo Laredo and Reynosa on the border, in Aguascalientes and Querétaro in the Bajío auto-electronics cluster, and in Saltillo in the northeast, all of which belong to the corridors of Figure 1. The central-eastern states, where European firms cluster, lose less than the national average, because foreign plants there hold a smaller share of tradable revenue and tradables pay a smaller share of the wage bill. Figure 5 plots the same results as income-weighted distributions across CZs. Because the common price term moves both densities to the left, no CZ gains under the all-exporter tariff. More than two-thirds of the CZs lose less than 0.02, but together they hold only a third of national income, so the loss falls on the CZs that hold most of the income. Under the all-exporter

tariff, 63 of the 738 panel CZs fall below -0.10, 44 of which are in the heavy-foreign group, whereas 17 do so under the foreign-only tariff.

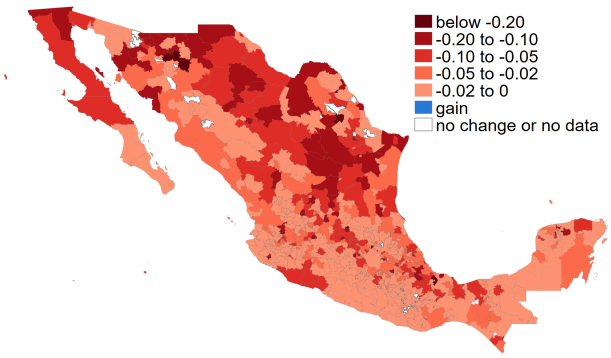
What the local labor market does to the incidence. Every employer in a commuting zone hires at the same wage, so a tariff that only reaches tradables moves the wage by the tradable revenue shock scaled by λ_c, the tradable share of local labor payments (equation (26)), which I measure as the tradable share of the CZ wage bill in the census. This has two implications for reading the maps. First, the pass-through from tradable revenue to local wages is well below one, and it varies across space with λ_c. A border CZ full of exporters has a much larger share of its wage bill exposed than an interior CZ that is mostly services, so the same proportional tariff shock moves the wage more at the border. The spatial gradient in welfare thus depends on two separate margins. One is how much the CZ's tradable revenue falls, $\widehat{R}_{\mathcal{T}c}$, which depends on the export shares of its tradable cells, and the other is how much of the local wage bill these tradables pay, λ_c. The two do not have to go together, since a few heavily taxed exporters can be located in a large services economy, and a large tradable sector may export little. Second, employment moves across sectors instead of disappearing. Non-tradable employers hire the workers that tradables let go at the new common wage, so total local employment stays exactly the same (Appendix E.2, Step 5). The welfare loss is therefore a fall in the price of labor and not a rise in local unemployment, which the fixed local labor supply rules out by construction.

5.4 Isolating the Agglomeration Channel

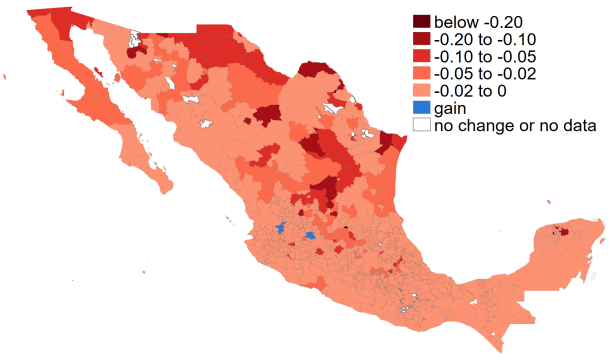
This subsection isolates how much of the welfare loss is due to the agglomeration feedback of Mechanism 2 (equation (25)) rather than to the direct incidence of Mechanism 1 (equations (22)–(24)) and the wage response of Mechanism 3 (equation (26)).

Design. In the counterfactual equilibrium, which is the exact-hat fixed point of Appendix E.2, the tariff only enters through the export-return factor and the export share of cell revenue. From there it spreads through three forces. These are entry and relocation (Mechanism 1), the agglomeration feedback (Mechanism 2), and the loop between wages and input prices of Mechanism 3, in which cell revenue moves the market-clearing wage and the wage moves $\tilde{P}_{jc}$, which enters both entry equations again. Since agglomeration depends on firm mass, a run that freezes the masses also has $\widehat{T_{ojc}} = 1$, which leaves only one way to nest the runs. All runs use $\kappa = 0.25$. In the *direct* run I freeze the wage and all firm masses, so that only the mechanical loss of export revenue is left. The *wage* run lets the wage and the input bundle adjust. The *mass* run also lets entry and relocation adjust, keeping $\widehat{T_{ojc}} = 1$, and the full

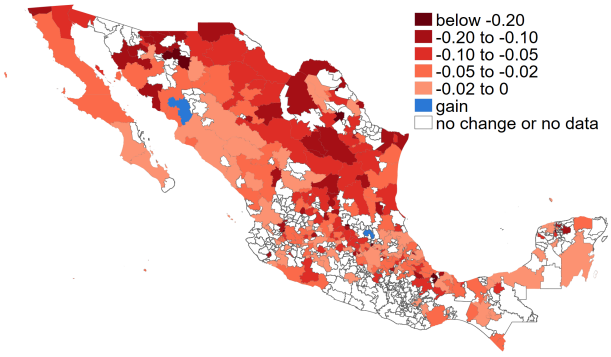
Figure 4: CZ-Level Welfare and Income Change under Each Tariff Scenario



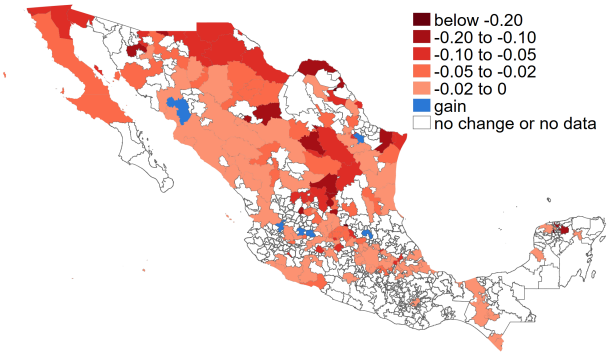
(a) Welfare, all-exporter tariff



(b) Welfare, foreign-only tariff

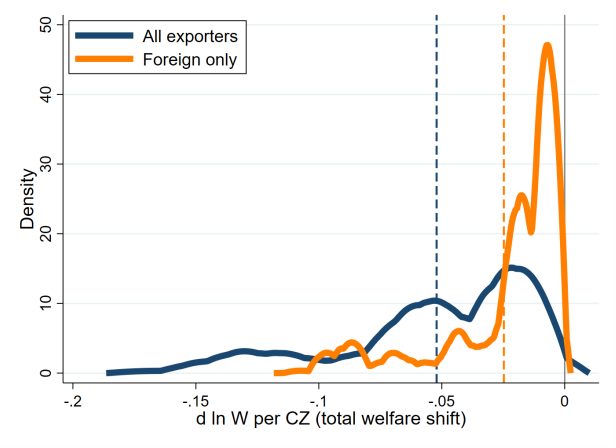


(c) Income, all-exporter tariff

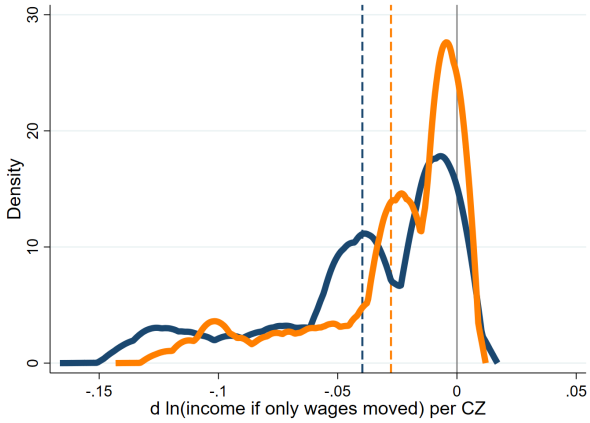


(d) Income, foreign-only tariff

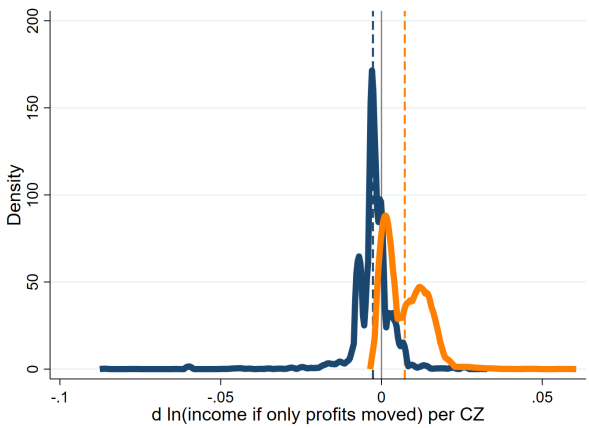
Figure 5: CZ Distribution of Welfare Shift and Income Sub-Channels



Total welfare shift



Wages sub-channel



Domestic profits sub-channel

Note: Dashed vertical lines mark the mean response.

run adds agglomeration. Let Δ_c^D denote the welfare change of the direct run, CZ by CZ, and Δ_c^w, Δ_c^M and Δ_c^T the change from each run to the next, so that the four terms sum exactly to the total welfare change:

$$d\ln W_c = \underbrace{\Delta_c^D}_{\text{direct}} + \underbrace{\Delta_c^w}_{\text{wage}\,|\,D} + \underbrace{\Delta_c^M}_{\text{mass}\,|\,D,w} + \underbrace{\Delta_c^T}_{\text{agglom.}\,|\,D,w,M} .$$

Note that each term includes its interactions with the channels that are already active. The agglomeration term Δ_c^T is therefore the feedback once wages and masses have adjusted, and it should not be read as a marginal effect free of interactions. Table 7 reports the four terms.

Table 7: Mechanism Chain: Sequential Decomposition of the Welfare Change

Term	All-exporter mean (sd)	Foreign-only mean (sd)
Direct (Δ_c^D: tariff alone, wages and masses frozen)	−0.011 (0.019)	0.000 (0.000)
Wage loop ($\Delta_c^w \mid D$)	−0.039 (0.048)	−0.026 (0.036)
Mass ($\Delta_c^M \mid D, w$)	−0.002 (0.020)	0.003 (0.015)
Agglomeration ($\Delta_c^T \mid D, w, M$)	−0.000 (0.009)	−0.002 (0.008)
Total $d\ln W_c$	−0.052 (0.043)	−0.025 (0.028)

Note: $\kappa = 0.25$, income-weighted means and unweighted standard deviations, untrimmed matched panel (738 CZs), so the means sum to the total.

The chain locates the largest single part of the loss in the wage. The direct term (−0.011 under the all-exporter tariff) is the mechanical export-revenue loss of Mexican-owned exporters; it is nil under foreign-only, since foreign profits do not accrue to Mexican households. The wage loop is the largest term in both scenarios, −0.039 and −0.026: the non-tradable sector in the labor-market clearing condition damps the wage response but does not remove it. Releasing entry and relocation nets to almost nothing (−0.002 under the all-exporter tariff and +0.003 under foreign-only), because it moves income and prices in opposite directions: the lower wage draws Mexican-owned entrants in (equation (22)), which raises labor demand and domestic profits, so income rises by 0.015 and 0.012, while the plants that close lower variety and raise the price index by about as much (0.017 and 0.009). The agglomeration feedback is close to zero, under 0.001 under the all-exporter tariff and 0.002 under foreign-only: weighted by domestic sales, the erosion of exporters' stocks barely moves productive capacity ($d\ln T_c$ averages 0.000, Table 6).

6 Conclusion

This paper asks how foreign exporting firms sort and cluster across space, and that clustering translates into welfare costs when a trade shock hits. Foreign firms in Mexico are scattered across several such corridors, and their location patterns differ sharply by country of origin. I rationalize this dispersion with an entry model in which foreign firms draw their productivity in every CZ and locate where it pays most, and each foreign origin-industry pair carries its own agglomeration primitive, estimated from establishment-level entry counts by Possion estimations. The estimated agglomeration force runs through origin nd industry channels. Fed into an exact-hat counterfactual, a 25% tariff on all Mexican manufacturing exports lowers income in the income-weighted average CZ by 0.042 log points (SD 0.033), nearly all of it through wages, and a higher tradable price index adds 0.010; restricting the tariff to foreign-owned exporters leaves over two-fifths of the income loss in place. The baseline geography of foreign exporters, differentiated by origin, determines where these losses land. A single tariff reaches every corridor at once, and the loss is largest where foreign exporters hold more of tradable revenue and tradables pay more of the local wage bill: along the northern border and in Bajío centers such as Aguascalientes and Querétaro.

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Appendix

A Input Prices: Wage Variation

Only the wage component of the input bundle varies across commuting zones in the estimation, so this appendix records just what the wage-based identification uses: the Cobb-Douglas unit cost, the labor-market-clearing wage, and the empirical reduction of $\tilde{P}_{jc}$ to the wage.

Unit cost. Under the Cobb-Douglas technology $y_i = \phi_o z_{ojc} L_i^{\beta_{Lj}} M_i^{\beta_{Mj}} E_i^{\beta_{Ej}}$ with inputs indexed by $k \in \{L, M, E\}$ and $\sum_k \beta_{kj} = 1$, cost minimization delivers marginal cost $MC_{jc} = \tilde{P}_{jc}/(\phi_o z_{ojc})$ with $\tilde{P}_{jc} \equiv \prod_k (p_{kc})^{\beta_{kj}}$, recovering the bundle posited in equation (2). The constant $\prod_k \beta_{kj}^{-\beta_{kj}}$ varies only by industry and year; I normalize it into productivity, which rescales T_{ojc} by a factor the fixed effects absorb. Cost minimization also implies constant cost shares: spending on input k is $\beta_{kj} VC_i$, with VC_i the firm's variable cost.

Wage determination. Under CES demand and monopolistic competition, variable cost is a fixed share of revenue, $VC_i = \frac{\sigma-1}{\sigma} r_i$, so labor expenditure is $w_c L_i = \frac{\sigma-1}{\sigma} \beta_{Lj} r_i$. Non-tradable producers are competitive, with a constant-returns Cobb-Douglas technology, so they price at unit cost, earn zero profits and pay labor the share β_N of their revenue, $w_c L_{Nc} = \beta_N R_{Nc}$. The wage hat weights revenue changes by wage bills, which are observed, so it needs neither factor. Aggregating across firms within (j, c), adding the non-tradable sector as a single industry with revenue R_{Nc}, and using intra-CZ labor mobility (a common w_c) with market clearing $\sum_{j \in \mathcal{T}} L_{jc} + L_{Nc} = L_c$ delivers the wage (11). The non-tradable sector has no entry margin and no productivity draws; it enters the model only through this condition.

Only the wage varies spatially. Capital is mobile within Mexico at a common cost of credit; electricity is administered by the federal electricity utility (CFE) at common rates and natural-gas and fuel pricing was government-controlled over most of the sample; and tradable materials are priced in national or international markets. Local advantages in access to materials, capital, or energy instead load into the productivity scale T_{ojc}. Hence

$$\ln \tilde{P}_{jct} \;=\; \beta_{Lj,t} \ln w_{c,t-5} \;+\; \alpha_{jt}, \tag{27}$$

where α_{jt} collects the non-labor components, varies only at the industry-year level, and is absorbed by the fixed effects (the origin-industry-year effect in the foreign equation, the industry-year effect in the domestic one). Scaled by $-\hat{\theta}$, the term $\beta_{Lj,t} \ln w_{c,t-5}$ is the wage term in the offset of both entry equations (Section 4.2), with $\beta_{Lj,t}$ the industry-year labor

share of variable cost. In the counterfactual, $p_{M,t}$ and $p_{E,t}$ stay at baseline and tariff shocks propagate to $\tilde{P}_{jct}$ entirely through equilibrium wages.

B Derivation of the Domestic Entry Equation

This appendix derives the coefficient on $\ln \tilde{P}_{jct}$ in the domestic Poisson intensity (18) from the zero-profit cutoff under a constant fixed cost and the Poisson (rare-event) approximation to the Fréchet survival probability.

Setup. A domestic firm in industry j at CZ c draws productivity z from a Fréchet distribution with shape parameter θ and scale T_{djc}:

$$F(z) \;=\; \exp\big(-T_{djc}\, z^{-\theta}\big), \qquad z > 0.$$

Core productivity ϕ_d is drawn from G_d independently of z (Section 3); Steps 1–4 condition on ϕ_d, and Step 5 integrates it out. Marginal cost of a firm with productivity z is $\mathrm{MC}(z) = \tilde{P}_{jc}/(\phi_d\, z)$, where $\tilde{P}_{jc}$ is the local input-price bundle (Appendix A). Under CES demand with elasticity $\sigma > 1$, monopolistic-competition markup pricing gives $p(z) = (\sigma/(\sigma-1))\,\mathrm{MC}(z)$. Industry-$j$ demand for a variety at price p is $q(p) = Y_j^d\,(P_j^d)^{\sigma-1}\,p^{-\sigma}$, where $Y_j^d \equiv \chi\alpha_j Y_d$ is domestic expenditure on industry j and P_j^d is the industry CES price index (3). The per-period fixed cost f_c^d of serving the domestic market is a constant (in numeraire units), not proportional to $\tilde{P}_{jc}$.

Step 1: firm revenue. Combining price and quantity:

$$\begin{aligned} r(z) \;&=\; p(z)\,q(p(z)) \;=\; Y_j^d\,(P_j^d)^{\sigma-1}\,p(z)^{1-\sigma} \\ &=\; Y_j^d\,(P_j^d)^{\sigma-1}\,\left(\tfrac{\sigma}{\sigma-1}\right)^{1-\sigma}\,\tilde{P}_{jc}^{1-\sigma}\,(\phi_d\, z)^{\sigma-1}. \end{aligned}$$

Collect terms that do not depend on z or on $\tilde{P}_{jc}$ into a constant A_j:

$$A_j \;\equiv\; Y_j^d\,(P_j^d)^{\sigma-1}\,\left(\tfrac{\sigma}{\sigma-1}\right)^{1-\sigma}\,\phi_d^{\sigma-1}, \qquad r(z) \;=\; A_j\,\tilde{P}_{jc}^{1-\sigma}\,z^{\sigma-1}.$$

Step 2: zero-profit cutoff. Operating profit under CES markup pricing is r/σ, so the zero-profit condition sets

$$\frac{r(z_{jc}^{d,*})}{\sigma} \;=\; f_c^d \qquad\Longleftrightarrow\qquad A_j\,\tilde{P}_{jc}^{1-\sigma}\,(z_{jc}^{d,*})^{\sigma-1} \;=\; \sigma\, f_c^d.$$

Isolate $(z_{jc}^{d,*})^{\sigma-1}$:

$$(z_{jc}^{d,*})^{\sigma-1} \;=\; \frac{\sigma\, f_c^d}{A_j}\, \tilde{P}_{jc}^{-(1-\sigma)} \;=\; \frac{\sigma\, f_c^d}{A_j}\, \tilde{P}_{jc}^{\sigma-1}.$$

Step 3: solve for the cutoff. Raise both sides to the power $1/(\sigma-1)$:

$$z_{jc}^{d,*} \;=\; \left(\frac{\sigma\, f_c^d}{A_j}\right)^{1/(\sigma-1)} \tilde{P}_{jc}. \tag{28}$$

A higher input-price bundle raises the cutoff one for one, $\partial \ln z_{jc}^{d,*} / \partial \ln \tilde{P}_{jc} = 1$.

Step 4: Poisson approximation to the survival probability. Under Fréchet productivity, the probability that a firm draws $z > z_{jc}^{d,*}$ is

$$\Pr(z > z_{jc}^{d,*}) \;=\; 1 - \exp\big(-T_{djc}\,(z_{jc}^{d,*})^{-\theta}\big).$$

In the rare-event limit where $T_{djc}\,(z_{jc}^{d,*})^{-\theta}$ is small (equivalently, the cutoff is far in the upper tail), the survival probability admits the Poisson approximation

$$\Pr(z > z_{jc}^{d,*}) \;\approx\; T_{djc}\,(z_{jc}^{d,*})^{-\theta}.$$

In that limit the Fréchet upper tail is asymptotically Pareto, the productivity distribution that Chaney (2008) assumes.

Step 5: expected entrant count. With M_{djct}^E potential domestic entrants in cell (d, j, c, t), the expected number of realized entrants is

$$\mathbb{E}[m_{djct}] \;=\; M_{djct}^E \cdot \Pr(z > z_{jc}^{d,*}) \;\approx\; M_{djct}^E\, T_{djc}\,(z_{jc}^{d,*})^{-\theta}.$$

Substituting the cutoff from (28):

$$\mathbb{E}[m_{djct}] \;\approx\; M_{djct}^E\, T_{djc} \left(\frac{\sigma\, f_c^d}{A_j}\right)^{-\theta/(\sigma-1)} \tilde{P}_{jc}^{-\theta}.$$

Since $A_j \propto \phi_d^{\sigma-1}$, the intensity is proportional to ϕ_d^θ; integrating over G_d replaces ϕ_d^θ with its mean $\mathbb{E}[\phi_d^\theta]$, which I assume finite. That mean is an industry-level constant absorbed by ν_{jt}, so the coefficient on $\ln \tilde{P}_{jc}$ is unchanged.

Step 6: log-linear form. Taking logs:

$$\begin{aligned}\ln \mathbb{E}[m_{djct}] \;=\; & \ln M^{E}_{djct} + \ln T_{djc} \\ & - \frac{\theta}{\sigma - 1} \ln\!\left(\frac{\sigma\, f^{d}_{c}}{A_j}\right) - \theta \,\ln \tilde{P}_{jc}.\end{aligned}$$

The first term, $\ln M^{E}_{djct}$, enters the offset. The second term, $\ln T_{djc}$, carries the same-origin and same-industry agglomeration loadings collected in $\beta' X_{djct}$. The third term is a function of industry-time constants (through A_j, which contains Y^{d}_{j}, P^{d}_{j}, σ, and, after integration, $\mathbb{E}[\phi_d^{\theta}]$) and of the fixed-cost requirement f^{d}_{c}; the industry-year fixed effect ν_{jt} absorbs the first and the CZ fixed effect ζ_c the second. The fourth term carries the coefficient $-\theta$.

The CES elasticity σ enters only through the constant A_j (via Y^{d}_{j}, P^{d}_{j}, and the markup factor), which is absorbed by fixed effects and does not appear in the identifying coefficient. Under a constant fixed cost, σ cannot be recovered from this equation and is identified separately from firm-level markups (Antràs et al., 2017); see Section 4.3.

Comparison with the foreign entry equation. The foreign intensity (17) carries the same $-\theta$ because the Fréchet draws raise the value's $\tilde{P}^{1-\sigma}_{jc}$ to $\theta/(\sigma-1)$ in the choice probability (16). In both equations θ is imposed from (20) through the offset.

C Details on Foreign Ownership

Table 8 reports mean foreign capital participation among establishments classified as foreign, by origin group and census wave. Two patterns stand out. Foreign ownership is concentrated near 100%: from 2004 onward every origin group averages between 78% and 96%, and the means for Asia and Europe rise across waves, from about 75% in 1994 to above 90% by 2024, consistent with the shift from minority joint ventures toward wholly owned subsidiaries after NAFTA. The 1994 and 1999 censuses record participation in a coded field and, separately, a flag for whether the establishment has foreign capital; in those waves an establishment is classified as foreign only if it also carries the flag, and for US and Canadian capital in 1999 the threshold is relaxed to the coded value 2. The 1994 figures are establishment-level means; the 1999–2024 figures are means over the (origin, industry, CZ) cells of the estimation panel.

D Derivation of the Hat Equations

The counterfactual is solved in relative changes, $\widehat{x} \equiv x'/x$, with x' the counterfactual value and x the baseline: each equilibrium condition is written under the tariff, divided by its

Table 8: Foreign Capital Participation (%) among Foreign-Owned Establishments, by Origin and Census Wave

Wave	North America	Asia	Europe	Other
1994	82.9	76.4	74.9	61.9
1999	84.4	83.2	86.3	73.8
2004	89.2	87.0	80.0	92.1
2009	89.4	88.6	84.3	92.4
2014	89.5	93.1	85.9	91.3
2019	78.3	95.1	93.0	89.9
2024	80.0	95.5	91.4	87.6

Note: Mean foreign capital participation among establishments classified foreign, by origin group. 1994: establishment-level mean; 1999–2024: mean over estimation-panel cells.

baseline version, and only the ratios are solved for (Caliendo and Parro, 2015, Dekle et al., 2008). Levels of the fundamentals cancel in the division, so the fixed point needs only the estimated elasticities and baseline objects observed in the data. This appendix derives the response equations of Section 5.2 from the entry equations of Section 4, and the price-index hats behind (21). Throughout, the foreign pools M^E_{ojt}, the birth rates b_{jt}, the exit rates δ_{jc}, the fixed effects and the parameters $(\theta, \sigma, \beta, \gamma)$ are held at their baseline values, and the tariff enters only through export revenue. The export propensity p^x_{ojc} is also held at baseline, abstracting from moves in the export cutoff and from the change in T_{ojc} that would shift the share of draws clearing it, and ω^x_{ojc} below is the observed baseline export share; γ therefore drops out, and the counterfactual omits marginal exporters that stop exporting. The domestic pool $M^E_{djct} = b_{jt} L^O_{jc,t-5}$, the industry's lagged employment in the CZ, is predetermined and is held at its baseline value as well.

Domestic firm mass, eq. (22). With the parameterization $T_{djc} = \exp(\beta' X_{djct})$, the domestic Poisson intensity (18) is

$$\mathbb{E}[m_{djct}] \;=\; b_{jt}\, L^O_{jc}\, T_{djc}\, \tilde{P}^{-\theta}_{jc}\, e^{\nu_{jt}+\zeta_c}.$$

Under the tariff T and $\tilde{P}$ change, while the pool stays at its baseline value:

$$\mathbb{E}[m'_{djct}] \;=\; b_{jt}\, L^O_{jc}\, T'_{djc}\, \tilde{P}'^{-\theta}_{jc}\, e^{\nu_{jt}+\zeta_c}.$$

Dividing the second line by the first, the pool and the fixed effects cancel,

$$\widehat{m}_{djc} \;=\; \frac{T'_{djc}}{T_{djc}}\left(\frac{\tilde{P}'_{jc}}{\tilde{P}_{jc}}\right)^{-\theta} \;=\; \widehat{T_{djc}}\,\widehat{\tilde{P}}_{jc}^{\,-\theta}.$$

Firms exit at the exogenous rate δ_{jc}, so the mass of operating firms is entry over exit, $M^d_{jc} = m_{djc}/\delta_{jc}$; with δ_{jc} fixed, $\widehat{M^d_{jc}} = \widehat{m}_{djc}\,\widehat{\varepsilon}_{djc}$, which is (22): because the cutoff is set on total profit, the export-return factor $\widehat{\varepsilon}_{djc}$ (derived below for foreign cells) also enters, although the estimated domestic intensity does not yet carry the export terms. The fixed effect ν_{jt} contains the industry demand terms Y^d_j and P^d_j of Appendix B, which also move under the tariff; holding them fixed drops a shift in domestic entry that is common to every CZ in the industry and does not affect where entry moves.

Foreign firm mass, eq. (23). The numerator of (16) is the right-hand side of (9) without ϕ_o, written here with its export factor separated,

$$S_{ojc} \;\equiv\; \delta_{jc}^{-\frac{\theta}{\sigma-1}}\,T_{ojc}\,\tilde{P}_{jc}^{-\theta}\,\varepsilon_{ojc}, \qquad \varepsilon_{ojc} \;\equiv\; \left(1 + p^x_{ojc}\,\tau^{x,1-\sigma}_{jc}\right)^{\frac{\theta}{\sigma-1}},$$

so that $\mathbb{P}(c \mid ojt) = S_{ojc}/\sum_{c'} S_{ojc'}$ and, by (17), $\mathbb{E}[m_{ojct}] = M^E_{ojt}\,\mathbb{P}(c \mid ojt)$. Under the tariff,

$$\widehat{m}_{ojc} \;=\; \frac{\mathbb{P}'(c \mid ojt)}{\mathbb{P}(c \mid ojt)} \;=\; \frac{S'_{ojc}/S_{ojc}}{\sum_{c'} S'_{ojc'}/\sum_{c'} S_{ojc'}}.$$

The numerator ratio is $\widehat{S}_{ojc} = \widehat{T_{ojc}}\,\widehat{\tilde{P}}_{jc}^{\,-\theta}\,\widehat{\varepsilon}_{ojc}$, with $\widehat{\varepsilon}_{ojc} \equiv \varepsilon'_{ojc}/\varepsilon_{ojc}$; the exit rate cancels. The denominator ratio is a baseline-share-weighted mean of the numerator ratios,

$$\frac{\sum_{c'} S'_{ojc'}}{\sum_{c'} S_{ojc'}} \;=\; \sum_{c'} \frac{S_{ojc'}}{\sum_{c''} S_{ojc''}}\,\widehat{S}_{ojc'} \;=\; \sum_{c'} \mathbb{P}(c' \mid ojt)\,\widehat{S}_{ojc'},$$

so $\widehat{\mathbb{P}}(c \mid ojt) = \widehat{S}_{ojc}/\sum_{c'} \mathbb{P}(c' \mid ojt)\,\widehat{S}_{ojc'}$, which is (23). The operating mass is the entrants that clear the cutoff over exit, $M^f_{ojc} = m_{ojc}\,\mathbb{P}(z_{ojc} \geq z^{f,*}_{ojc} \mid c)/\delta_{jc}$. A plant in c holds the winning draw, whose value, variable profit over the exit rate, is Fréchet with shape $\theta/(\sigma-1)$ and scale proportional to $\sum_{c'} S_{ojc'}$ whichever CZ won; it clears the cutoff when that value exceeds its fixed costs over the exit rate, which do not move in the counterfactual. Under the rare-event approximation of Appendix B the surviving share is therefore proportional to $\sum_{c'} S_{ojc'}$ times a term in the fixed costs and exit rate of c, and since the hat of that sum is the denominator above, $\widehat{M^f_{ojc}} = \widehat{\mathbb{P}}(c \mid ojt)\sum_{c'} \mathbb{P}(c' \mid ojt)\,\widehat{S}_{ojc'} = \widehat{T_{ojc}}\,\widehat{\tilde{P}}_{jc}^{\,-\theta}\,\widehat{\varepsilon}_{ojc}$, which is (24): entrants redistribute across CZs, and fewer of them operate when the tariff lowers the pair's

total attractiveness. The baseline shares $\mathbb{P}(c \mid ojt)$ are the fitted counts of the foreign PPML, normalized within each origin-industry-year.

The export-return factor. A plant's revenue is its domestic revenue times $1+p^x_{ojc}\tau^{x,1-\sigma}_{jc}$. The tariff multiplies export revenue by $(1+\kappa)^{1-\sigma}$, so the ratio of revenue after to before is

$$\frac{1+p^x_{ojc}\,\tau^{x,1-\sigma}_{jc}(1+\kappa)^{1-\sigma}}{1+p^x_{ojc}\,\tau^{x,1-\sigma}_{jc}} \;=\; (1-\omega^x_{ojc})+\omega^x_{ojc}\,(1+\kappa)^{1-\sigma}, \qquad \omega^x_{ojc} \;\equiv\; \frac{p^x_{ojc}\,\tau^{x,1-\sigma}_{jc}}{1+p^x_{ojc}\,\tau^{x,1-\sigma}_{jc}},$$

where ω^x_{ojc} is the export share of revenue, the object the census reports; it is the regressor of Table 3. Raising the ratio to $\theta/(\sigma-1)$ gives $\widehat{\varepsilon}_{ojc}$ as written in (23), without any need for τ^x_{jc} itself. When masses move, the cell revenue hat used below is the mass hat, $\widehat{R}_{ojc}=\widehat{M_{ojc}}$: under the rare-event approximation, the average revenue of the plants that clear the cutoff is a fixed multiple of the fixed cost, so the export loss reaches cell revenue only through the number of plants that survive, which $\widehat{\varepsilon}_{ojc}$ already carries, and multiplying the mass hat by the revenue ratio as well would count the loss twice. In the runs of Section 5.4 that freeze masses, every plant keeps operating and its revenue falls by the revenue ratio, so $\widehat{R}_{ojc}$ is the ratio itself. Under the foreign-only tariff, $\widehat{\varepsilon}_{ojc}=1$ and the revenue ratio is one for Mexican-owned cells.

Price index. By (12) the industry index is the product of a mass term and a productive-capacity term, so

$$\widehat{P^d_j} \;=\; \widehat{M}_j^{\frac{1}{1-\sigma}}\,\widehat{TP}_j^{-\frac{1}{\theta}}, \qquad \widehat{TP}_j \;\equiv\; \sum_c \omega_{jc}\,\widehat{T_{jc}}\,\widehat{\widetilde{P}}_{jc}^{\,-\theta},$$

where $\widehat{M}_j$ is the baseline-revenue-weighted mean of the cell mass hats of (22) and (24), ω_{jc} are the productive-capacity weights of Section 5.1, measured by baseline domestic sales, and $\widehat{T_{jc}}=\sum_o T_{ojc}\widehat{T_{ojc}}/\sum_o T_{ojc}$ is the hat of the origin-aggregated scale, whose origin weights are measured the same way. Taking logs gives the variety and the productivity and input-price terms of (21), which add up to the price channel. Inside the second term, the cell changes $\ln\widehat{T_{jc}}$ and $\ln\widehat{\widetilde{P}}_{jc}$ are weighted by the logarithmic mean of their baseline and counterfactual capacity, so that the productivity and input-price parts add up to the term exactly.

Productivity, eq. (25). With the parameterization of Section 4 (foreign loadings for foreign cells, domestic loadings for Mexican cells), the difference between the counterfactual and

the baseline is

$$\ln \widehat{T_{ojc}} \;=\; \beta_{og} \ln \frac{1+(wL^{o,c})'}{1+wL^{o,c}} + \beta_j \ln \frac{1+(wL^{j,c})'}{1+wL^{j,c}} \;\approx\; \beta_{og} \ln \widehat{wL^{o,c}} + \beta_j \ln \widehat{wL^{j,c}},$$

the approximation being close for all but the smallest stocks. Under Cobb-Douglas cost minimization each cell's wage bill is a fixed share of its revenue, so a cell's wage bill moves with $\widehat{R}_{ojc}$ and each stock moves by the baseline-weighted mean of its cells' revenue hats, as displayed in Section 5.2. Substituting gives (25).

Input bundle and wages. By (27) only the wage varies across CZs, so $\widehat{\tilde{P}}_{jc} = \widehat{w}_c^{\beta_{Lj}}$. Labor-market clearing (11) gives the wage hat (26) at fixed local labor supply and unchanged non-tradable revenue (Appendix E.2, Step 5). The mass hats, the stocks, the wage and the input bundle form the fixed point that Appendix E.2 iterates.

E Estimation and Counterfactual Algorithm

This appendix walks through the estimation and counterfactual procedures in the order they are executed. Section E.1 covers the recovery of σ and θ and PPML estimation. Section E.2 covers the exact-hat fixed point and welfare-decomposition aggregation used in Section 5.

E.1 Estimation

1. **Recover $\hat{\sigma}$ from firm-level markups** as in Section 4.2 (median markup over labor, materials and energy, inverted wave by wave).

2. **Recover $\hat{\theta}$ from the size distribution** by the moment inversion (20) of Section 4.2, and check that $\hat{\sigma} < \hat{\theta} + 1$ (integrability).

3. **Foreign PPML** (17). Estimate by Poisson pseudo-maximum likelihood absorbing a fully saturated origin-industry-year (ν_{ojt}) fixed effect, with $\ln M^E_{ojt} - \hat{\theta}\,\beta_{Lj,t} \ln w_{c,t-5} + \frac{\hat{\theta}}{\hat{\sigma}-1} \ln \delta^{-1}_{jc}$ in the offset. Standard errors clustered at the CZ level; singleton and separated observations within FE cells are dropped. The coefficients on X_{ojct} identify the agglomeration loadings $\hat{\beta}$; the coefficients on Z_{ojct} identify the export-margin loadings scaled by $\theta/(\sigma-1)$, $\frac{\hat{\theta}}{\hat{\sigma}-1}\hat{\gamma}$, from which $\hat{\gamma}$ follows given $\hat{\theta}$ and $\hat{\sigma}$.

4. **Domestic PPML** (18). Estimate by PPML absorbing industry-year (ν_{jt}) and CZ (ζ_c) fixed effects, with $\ln M^E_{djct} - \hat{\theta}\,\beta_{Lj,t} \ln w_{c,t-5}$ in the offset. Singleton and separated observations within FE cells are dropped. The coefficients on X_{djct} identify the domestic agglomeration loadings.

E.2 Counterfactual: Exact-Hat Algebra

For each scenario and tariff rate of Section 5.1, the simulator iterates the hat equations of Appendix D to a fixed point.

1. **Choice probability hat**. $\widehat{\mathbb{P}}(c \mid ojt)$ from (23), with $\widehat{\varepsilon}_{ojc}$ as defined in Section 5.2.

2. **Firm mass hat**. $\widehat{M^f_{ojc}}$ from (24) for foreign cells (origin-industry entrant pool fixed at baseline); $\widehat{M^d_{jc}} = \widehat{T_{djc}}\,\widehat{\tilde{P}}_{jc}^{\,-\theta}\,\widehat{\varepsilon}_{djc}$ from (22) for Mexican cells (domestic entrant pool fixed at baseline), with $\hat{\theta}$ from Step 2 of Section E.1.

3. **Cell revenue hat**. $\widehat{R}_{ojc} = \widehat{M_{ojc}}$ (Appendix D); in the runs that freeze masses, $\widehat{R}_{ojc}$ is the export-revenue ratio.

4. **Agglomeration feedback**. Update $\widehat{T_{ojc}}$ through (25), with the stock hats of Section 5.2, the foreign loadings for foreign cells and the domestic loadings for Mexican cells; the update is damped by half a step, which leaves the fixed point unchanged.

5. **Wage clearing**. From (11) at fixed CZ labor supply, over both sectors,

$$\widehat{w}_c \;=\; \frac{\sum_{j\in\mathcal{T}} \frac{\sigma-1}{\sigma}\beta_{Lj,t}\, R_{jc}\, \widehat{R}_{jc} \;+\; \beta_N R_{Nc}\, \widehat{R}_{Nc}}{\sum_{j\in\mathcal{T}} \frac{\sigma-1}{\sigma}\beta_{Lj,t}\, R_{jc} \;+\; \beta_N R_{Nc}}.$$

   The tariff is levied on tradable exports and does not reach non-tradable output, so $\widehat{R}_{Nc} = 1$; with the non-tradable good as numeraire this is a statement about quantities, namely that non-tradable output is unchanged. The non-tradable term is then common to numerator and denominator. Writing λ_c for the tradable share of local labor payments,

$$\lambda_c \;\equiv\; \frac{\sum_{j\in\mathcal{T}} \frac{\sigma-1}{\sigma}\beta_{Lj,t}\, R_{jc}}{\sum_{j\in\mathcal{T}} \frac{\sigma-1}{\sigma}\beta_{Lj,t}\, R_{jc} + \beta_N R_{Nc}},$$

   this is the wage hat (26),

$$\widehat{w}_c \;=\; \lambda_c\, \widehat{R}_{\mathcal{T}c} \;+\; (1-\lambda_c), \qquad \widehat{R}_{\mathcal{T}c} \equiv \frac{\sum_{j\in\mathcal{T}} \beta_{Lj,t} R_{jc} \widehat{R}_{jc}}{\sum_{j\in\mathcal{T}} \beta_{Lj,t} R_{jc}},$$

   so the local wage moves by the tradable share of the wage bill times the tradable revenue shock. In the model labor payments are the wage bill, so λ_c and the weights in $\widehat{R}_{\mathcal{T}c}$ are measured directly from the observed CZ and cell wage bills. Sector employment follows from $L_{jc} = \frac{\sigma-1}{\sigma}\beta_{Lj} R_{jc}/w_c$: tradable employment moves by $\widehat{L}_{\mathcal{T}c} = \widehat{R}_{\mathcal{T}c}/\widehat{w}_c$ and non-tradable employment by $\widehat{L}_{Nc} = 1/\widehat{w}_c$. Because $\lambda_c \widehat{L}_{\mathcal{T}c} + (1-\lambda_c)\widehat{L}_{Nc} = \widehat{w}_c/\widehat{w}_c = 1$,

total local employment is conserved exactly: the workers tradables shed are hired by non-tradables at the new common wage. This is market clearing, not a damping factor applied after the fact.

6. **Input-price hat**. $\widehat{\tilde{P}}_{jc} = \widehat{w}_c^{\beta_{Lj,t}}$, under the input-price assumption of Appendix A.

7. **Iterate**. Repeat Steps 1–6 until the largest change in $\widehat{w}_c$ across CZs and in $\widehat{T_{ojc}}$ across cells is below 10^{-6} (at most 300 iterations).

8. **Assemble the welfare decomposition per CZ** following (21):

   - *Income channel.* $\ln \widehat{Y}_c = \ln(s_{w,c}\widehat{w}_c + s_{\pi,c}\widehat{\Pi}_c^d)$, where dividends are operating profits of Mexican-owned cells, $\Pi_c^d = \sum_j R_{djc}/\sigma$ gross of fixed costs, so that $\widehat{\Pi}_c^d = \sum_j R_{djc}\widehat{R}_{djc} / \sum_j R_{djc}$, and $(s_{w,c}, s_{\pi,c})$ are the baseline income shares. The wage and profit sub-channels are the income change when only wages or only profits move, $\ln(s_{w,c}\widehat{w}_c + s_{\pi,c})$ and $\ln(s_{w,c} + s_{\pi,c}\widehat{\Pi}_c^d)$, which sum to $\ln \widehat{Y}_c$ to first order.
   - *Price channel.* $\text{variety}_c = \chi\, \alpha_j \ln \widehat{M}_j/(\sigma - 1)$ and $\text{prod+input}_c = \chi\, \alpha_j \ln \widehat{TP}_j/\theta$, summed over industries, where $\widehat{M}_j$ is the baseline-revenue-weighted mean of the cell mass hats; both are constant across CZs (the price index is CZ-invariant under costless domestic trade). The second is split into its productivity and input-price parts as in Appendix D. The CZ-local block of Table 5 recomputes the same sub-channels under a CZ-specific aggregator restricted to the industries each CZ hosts, as a diagnostic of cross-CZ dispersion.

9. **Aggregate to Table 5** by taking the income-weighted mean of each CZ-level channel using baseline CZ income $w_c L_c + \Pi_c^d$. In this aggregation, outside the simulator, trim the welfare and wage channels at the 2nd/98th percentile and the profit and CZ-local price channels at the 0.5th/99.5th percentile within each scenario, so that a few extreme CZs do not dominate the moments.